Multispatial Filtering Module Cascaded System for Hyperspectral Image Classification

Xiaodi Shang, *Student Member, IEEE*, Meiping Song, *Member, IEEE*, and Chein-I Chang, *Life Fellow, IEEE*

***Abstract*— This article presents a multispatial filtering module cascaded system (MSFMCS) for hyperspectral image classification (HSIC), which can serve as a paradigm to improve spectral–spatial classification. It includes multiple spatial filtering modules (SFMs) that are cascaded to particularly capture spatial information from the classification maps generated from the preceding modules. As a result, any spectral classifier (SC) can be used as an input to an initial/input module (IM). Through MSFMCS, its classification performance keeps improving as more SFMs are processed. To terminate MSFMCS, an automatic stopping rule is particularly designed by support vector machine (SVM) which is used not only as a classifier but also as a decision-maker. So, once an SC cannot be further improved, MSFMCS is terminated. One major benefit resulting from MSFMCS is its framework which can implement any arbitrary SC as its initial classifier in IM. Another is its ability in capturing additional spatial classification information module by module as the process progresses. A third one is no weights connected between modules so that no training phase is required like a feedforward neural network. Finally, the number of modules used in MSFMCS can be automatically determined by its stopping rule not predetermined empirically. To illustrate full advantages of MSFMCS in HSIC, three types of heterogeneous classifiers, pure-pixel-based SVM, mixed-pixel-based constrained energy minimization (CEM), and feature-extraction-based classifier—orthogonal total variation component analysis (OTVCA)—are used for experiments to demonstrate how MSFMCS can improve their classification performance.**

***Index Terms*— Constrained energy minimization (CEM), edge preserving filter (EPF), fixed training sampling (FTS), multispatial filtering module cascaded system (MSFMCS), orthogonal total variation component analysis (OTVCA), random training sampling (RTS), spatial filtering module (SFM), support vector machine (SVM).**

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Xiaodi Shang and Meiping Song are with the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Information and Technology College, Dalian Maritime University, Dalian 116026, China (e-mail: shangxiaodi@qq.com; smping@163.com).

Chein-I Chang is with the Remote Sensing Signal and Image Processing Laboratory, Department of Computer Science and Electrical Engineering, University of Maryland, Baltimore County, Baltimore, MD 21250 USA (e-mail: cchang@umbc.edu).

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NOMENCLATURE

AA	Average accuracy.
APR	Average precision.
AVIRIS	Airborne visible/infrared imaging spectrometer.
BKG	Background.
MSFMCS	Multispatial filtering module cascaded system.
EPF	Edge preserving filter.
EPF-G-g	Guided filter using the first principal component as its reference image.
FTS	Fixed training sampling.
HSIC	Hyperspectral image classification.
IM	Initial/input module.
MAP	Maximum *a posteriori.*
OA	Overall accuracy.
OPR	Overall precision.
OTVCA	Orthogonal total variation component analysis.
PR	Precision rate.
RTS	Random training sampling.
SC	Spectral classifier.
SF	Spatial filter.
SFM	Spatial filtering module.
SCMaps	Spectral-classified maps.
SFMaps	Spectral-filtered maps
SVM	Support vector machine.
SVMaps	SVM-classified maps.

I. INTRODUCTION

HSIC has received considerable interest in recent years. Many techniques have been developed for HSIC and reported in the literature. One of the most widely used classifiers is the SVM [1]. Unfortunately, SVM is a pixel-wise SC which does not take the spatial correlation among pixels into account. To improve the SVM performance in classification, spatial classification information has been included in SVM. So, an earlier attempt was to introduce segmentation maps obtained by partitional clustering using majority voting [2]. Another interesting approach was developed in [3], called composite kernel (CK)-based method, which used spatial windows to extract spatial information directly from the original hyperspectral data to capture both spectral and spatial information. This approach was further improved by region-kernel-based SVM (RKSVM) to capture describe joint spatial–spectral similarity [4]. As an alternative to [2]–[4], Tarabalka *et al.* [5] used segmentation to obtain spatial

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information by Markov random field (MRF) regularization. Most recently, the spectral–spatial (SS)-based classification approach has emerged as one of the most promising HSIC methods. It first implements SVM as an SC followed by SFs to capture spatial information among SVM-classified data samples. As a result, many SS-based methods have been developed by implementing an SC in conjunction with a set of SFs to extract spatial-pixel information to improve classification [6]–[14].

As opposed to the SS classification methods in [1]–[14], a recent series of works, called iterative SS (ISS) classification techniques, were developed in [15]–[19]. These ISS classifiers extended SVM used in [1]–[5] and EPF in [7], [12], [14], and [15] to iterative versions so as to improve the classification performance. Interestingly, these ISS classifiers can also make subpixel detectors, constrained energy minimization (CEM) [15], [20]–[22], multiple-target detectors, and linearly constrained minimum variance (LCMV) [16], [22]–[24] to perform spectral classification. Since CEM and LCMV are originally designed as detectors to produce abundance fractions for targets to be detected, ISS classification allows CEM and LCMV to capture class spatial information via an iterative process to reduce uncertainty by increasing the detected abundance fractions of accuracy for classification.

Unlike SS-based methods which separate spectral classification and spatial filtering into two individual processes, many recent approaches have investigated in using models to learn spatial and spectral information jointly to improve classification such as deep learning neural networks (NNs) [25]–[29], convolutional neural network (CNN) [30]–[37], and generative adversarial network (GAN) [38]–[40], which have also been studied extensively in HSIC. For example, CNN uses a convolutional submodule to capture spatial pattern information from a visual receptive field and then subsamples the visual patterns via a pooling submodule. From this point of view, CNN can be considered to perform spatial/spectral classification. GAN requires generative and discriminative models to compete each other between fake/true image classification.

Compared with CNN and feedforward NNs [41], the SS classification does not take advantage of its multilayer feedforward architecture. So it raises an interesting question "Can we extend an SS classifier to a multilayer SS classifier similar to CNN?" This article investigates this issue and further develops a quite different but an innovative approach, called MSFMCS, which is not a familiar feedforward network as described above but rather extends SS classification to a new paradigm or HSIC in a forward cascade manner. There are several key differences in the designed MSFMCS from the existing HSIC techniques. First, MSFMCS does not use multiple layers as CNNs or deep learning networks do. Instead, it cascades SF-specified modules to form a system in a feedforward manner but not a multilayer feed forward network. Because the between-module connectors in MSFMCS are stopping rule as opposed to the between-layer connectors in the multilayer feed forward network which are weights needed to be trained. As a result, a second difference is that no training phase is required for MSFMCS to train the system. In particular, the key connecting the two modules is an SVM-based stopping rule not weights. Most importantly, MSFMCS designs two types of modules, an IM and subsequent SFMs. So, a third difference is that the MSFMCS designs an IM which can take any SC as its input and improves its classification performance as the MSFMCS process progresses. If an SC performs so well and cannot be further improved by subsequent SFMs, MSFMCS will be terminated in IM and is reduced to a single-module system which is exactly the original SC. Fourth, with the exception of IM, each of the subsequent SFMs implemented in MSFMCS comprises only a set of SFs deigned to capture class spatial information from its preceding module. Finally and most importantly, MSFMCS uses an automatic stopping rule to determine the number of modules, to be called length, to be cascaded.

The architecture of MSFMCS consists of four components. The first component is an IM considered as the input module, referred to as IM that inputs any SC to be improved by MSFMCS in classification. This is then followed by the second component, which is a cascade of SFMs, each of which implements a set of SFs to capture spatial class contextual information. The third component is an SVM-based stopping rule attached to the end of each SFM with its designed goal to determine whether the current SFM is needed to be continuously cascaded or terminated. Finally, the fourth component is the terminal SFM which is the last SFM in the system and outputs final classification results. Among the four components, it is worth noting that the SVM-based stopping rule implemented at the end of each of cascaded SFMs is the key connector to make the entire MSFMCS work.

Since training samples are generally required for SVM-based stopping rule, two different approaches for the selection of training samples developed in [14] are used. One is FTS which randomly selects an initial set of training samples in the input module. Then, this same set of training data is used throughout all SVM-based stopping rules between SFMs in MSFMCS. The other is RTS which is repeatedly used to select random training samples for each of all the subsequent SVMs implemented at the end of SFMs. As a result of RTS, all the training samples selected and used for each SFM are completely different and random. The same FTS and RTS can also be applied to the SC used in IM as well. Fig. 1 depicts a graphic block diagram of implementing MSFMCS at the kth SFM highlighted by the dot-dashed box. Specifically, to make MSFMCS in Fig. 1 simple to illustrate, the stopping rule is particularly specified by SVM where $\left\{\text{SVMap}_m^{(k)}\right\}_{m=1}^{M}$ is the output of the kth SFM resulting from the SVM classification using $\left\{\text{SVMap}_m^{(k-1)}\right\}_{m=1}^{M}$ as the input of the kth SFM, which is also the output from the $(k-1)$st SFM.

According to Fig. 1, MSFMCS initializes a system with implementing an SC in IM and then concatenates a set of multiple SFMs in a cascade to form a system. As noted, MSFMCS is not a multilayer feedforward NN in several aspects summarized in the following.

1) MSFMCS offers a new paradigm which provides a general platform to improve HSIC. Unlike many NN and deep-learning-based networks which require tuning many parameters to train networks, MSFMCS does not

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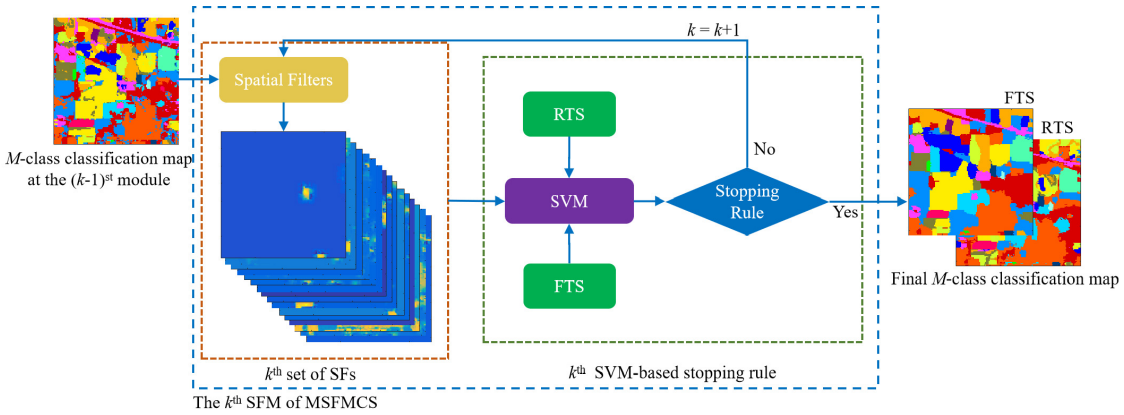


Fig. 1. Graphic diagram of implementing MSFMCS with the kth SFM.

need training data to train its system. So no parameters are needed.

2) MSFMCS uses its IM as the input module to take any SC as an input and keeps improving the input SC in its classification performance as MSFMCS progresses. MSFMCS will be terminated once the classification performance is not further improved.
3) MSFMCS considers each SFM as an independent module with its input being the output from its preceding SFM. Its system complexity is very simple and has a great benefit in hardware design.
4) MSFMCS concatenates two successive SFMs by an SVM classifier attached at the end of each SFM, which can serve two purposes. One is to determine whether the SC keeps improving its performance by MSFMCS. The other is to determine the number of needed SFMs used in MSFMCS.

The remainder of this article is organized as follows. Section II develops the framework of MSFMCS. Section III summarizes novelties of MSFMCS. Section IV derives the MSFMCS algorithm with a custom-designed automatic stopping rule. Section V focuses on allocation of training sample size for each class. Section VI describes images and classification measures to be used for study. Section VII analyzes the experimental results. Section VIII discusses the generalizability of MSFMCS to offer several future research directions. Finally, Section IX draws some conclusions.

II. Multispatial Filtering Module Cascaded System

A general framework of MSFMCS consists of SFMs concatenated one after another to form a cascaded system. It can be characterized as follows, IM as the input module and subsequent SFMs. The IM takes an SC as an input to the system and then the subsequent SFMs can be considered as independent modules which only take the output from the preceding module as its input. Any pair of two successive modules are connected by an SVM classifier which serves as a stopping rule. A final SFM, referred to as the terminal SFM, outputs the final classification results. In what follows, we describe each of these four elements of MSFMCS in great detail.

A. IM

IM is considered as an input module to initialize MSFMCS. It is different from its subsequent SFMs in the sense that it implements an SC without using a set of SFs, while all subsequent SFMs with $k \geq 1$ implement only a set of specified SFs, $\{\mathrm{SF}_m^{(k)}\}_{m=1}^{M}$, without SC as shown in Fig. 2(a) where M is the number of classes. With introducing SFMs into MSFMCS, many SCs can be reinterpreted as a special case of MSFMCS which implements only IM. For example, SVM and CEM alone can be considered as SC to be implemented in IM with a key difference. Since SVM itself is a classifier, it can produce classification maps by itself. To the contrary, CEM is a mixed-pixel classifier, and it requires a thresholding technique to perform classification. As a result, it needs a binarization technique such as MAP in [42] and Otsu's method [43] to produce classification maps.

B. Subsequent SFMs

The purpose of including additional new SFMs in MSFMCS is to resolve the issue arising in lack of spatial class information for SC to perform well. A kth SFM implements a set of M SFs, $\{\mathrm{SF}_m^{(k)}\}_{m=1}^{M}$, followed by an SVM-based stopping rule. The mth SF, $\mathrm{SF}_m^{(k)}$, is used to capture the spatial classification information for the mth class from the SVMaps, $\{\mathrm{SVMap}_m^{(k-1)}\}_{m=1}^{M}$, which will serve as the input to the $(k-1)$st SVM-based stopping rule. The SVM-based stopping rule is included as a decision-maker to determine whether or not a new SFM is needed to be cascaded at its end. Generally speaking, the used SFs in the kth SFM do not have to be the same SFs as other SFMs use and can vary with module by module.

Of particular interest is the first SFM. Since an SC alone generally does not take spatial information into account such as SVM and CEM, many classifiers recently developed for HSIC have included SFs to be implemented in conjunction with SC so as to improve the classification performance. In this case, these types of classifiers using SFs can be reinterpreted as special cases of the first SFM in MSFMCS with $k = 1$. As an example, EPF-based classifiers in [7] used SVM as an SC in IM to produce $\{\mathrm{SCMap}_m\}_{m=1}^{M}$

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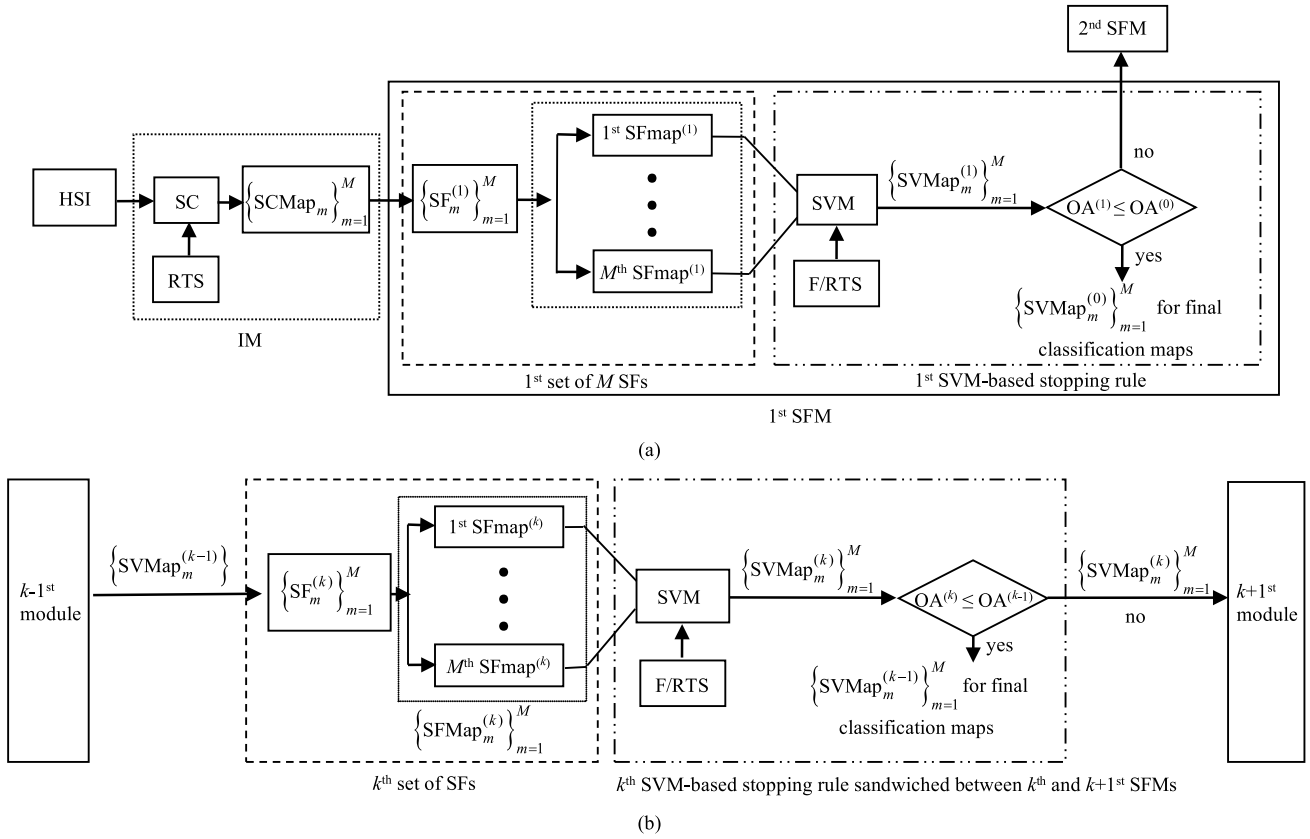


Fig. 2. Block diagrams of implementing MSFMCS. (a) IM of SC and first SFM consisting of M SFs, $\{\mathrm{SF}_m^{(1)}\}_{m=1}^{M}$, followed by the first SVM-based stopping rule. (b) kth SFM ($k \geq 2$).

denoted by $\{\mathrm{SVMap}_m^{(0)}\}_{m=1}^{M}$. It then used $\{\mathrm{SF}_m^{(1)}\}_{m=1}^{M}$ specified by EPF to generate $\{\mathrm{SFMap}_m^{(1)}\}_{m=1}^{M}$ which captured spatial class information among $\{\mathrm{SVMap}_m^{(0)}\}_{m=1}^{M}$. Finally, it used MAP to binarize $\{\mathrm{SFMap}_m^{(1)}\}_{m=1}^{M}$ to produce its final classification results. In this case, we can interpret EPF-based classifiers in the context of MSFMCS by replacing the SVM-based stopping rule used at the end of the first SFM with MAP. Accordingly, the EPF-based method in [7] becomes a special case of MSFMCS which uses only the first SFM. Similarly, if the SC in IM is specified by CEM, then $\{\mathrm{SCMap}_m\}_{m=1}^{M}$ becomes M CEM-detected classification maps, that is, $\{\mathrm{SCMap}_m\}_{m=1}^{M} = \{\mathrm{CEMap}_m^{(0)}\}_{m=1}^{M}$. Then $\{\mathrm{SF}_m^{(1)}\}_{m=1}^{M}$ are applied to $\{\mathrm{CEMap}_m^{(0)}\}_{m=1}^{M}$ to produce $\{\mathrm{SFMap}_m^{(1)}\}_{m=1}^{M}$. In this case, an SVM-based stopping rule will be further implemented at the end of the first SFM of MSFMCS. In particular, when the SFs and SVM-based stopping rule are replaced by Gaussian filters and Otsu's method [43], respectively, then the resulting first SFM in MSFMCS becomes CEM-based classifier derived in [15] which can also be considered as a special case of MSFMCS which implements only one single SFM as its first SFM classifier without further being cascaded.

C. SVM-Based Automatic Stopping Rule With $k \geq 2$

One of the key elements in MSFMCS is introduction of automatic stopping rule. It works as a connector between two successive SFMs. At the beginning of the kth SFM, it takes M SVM classification maps produced by the $(k-1)$st SVM-based automatic stopping rule at the end of $(k-1)$st SFM, denoted by $\{\mathrm{SVMap}_m^{(k-1)}\}_{m=1}^{M}$ as the input to $\{\mathrm{SF}_m^{(k)}\}_{m=1}^{M}$ in the kth SFM. Then, it takes the SVM-based stopping rule at the end of the kth SFM to perform spectral classification to produce $\{\mathrm{SVMap}_m^{(k)}\}_{m=1}^{M}$ where $\mathrm{SVMap}_m^{(k)}$ is actually a binary map for the classification map of the mth class, C_m. Then the OA is calculated for $\{\mathrm{SVMap}_m^{(k)}\}_{m=1}^{M}$ and $\{\mathrm{SVMap}_m^{(k-1)}\}_{m=1}^{M}$, denoted by $\mathrm{OA}(\{\mathrm{SVMap}_m^{(k-1)}\})$ and $\mathrm{OA}(\{\mathrm{SVMap}_m^{(k)}\})$. If

$$\mathrm{OA}\left(\{\mathrm{SVMap}_m^{(k)}\}\right) \leq \mathrm{OA}\left(\{\mathrm{SVMap}_m^{(k-1)}\}\right) \tag{1}$$

then MSFMCS is terminated and the current SFM becomes the output SFM. Otherwise, the results $\{\mathrm{SVMap}_m^{(k)}\}_{m=1}^{M}$ are passed on to the next $(k+1)$st SFM. As a result, the number of SFMs specified by length d can be automatically determined by (1).

D. Terminal SFM

Finally, when MSFMCS reaches the terminal SFM, it provides two significant pieces of information. One is the length of MSFMCS, d, which determines the number of SFMs to be cascaded in MSFMCS. The other is the results produced by the last SVM-based stopping rule, $\{\mathrm{SVMap}_m^{(d)}\}_{m=1}^{M}$, which can be used for final classification results. With this interpretation, the terminal SFM can also be referred to as the output SFM.

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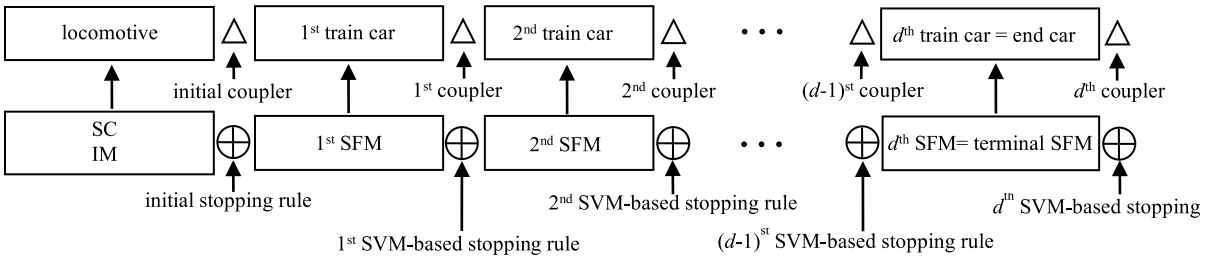


Fig. 3. MSFMCS illustrated by an example of using a train pulling a cascade of cars with couplers connected between two cars.

Fig. 2(a) depicts the block diagrams of MSFMCS at the IM implementing an SC and the first SFM implementing a set of SFs, $\{\mathrm{SF}_m^{(1)}\}_{m=1}^{M}$, along with its first SVM-based stopping rule. Fig. 2(b) describes the kth SFM with $k \geq 2$ which includes only a set of SFs, $\{\mathrm{SF}_m^{(k)}\}_{m=1}^{M}$, without an SC and its kth SVM-based stopping rule.

III. Novelties of MSFMCS

Intuitively speaking, MSFMCS is a simple system comprising a cascade of a number of SFMs with an SVM-based stopping rule sandwiched between two successive SFMs. Many current existing deep learning and NNs have model complexity with many parameters needed to be tuned to train networks. In contrast, MSFMCS requires neither a network model nor weights connecting between SFMs. Thus, no parameters are used in the system. Two novel ideas to make MSFMCS work are the development of IM and a new concept of introducing an automatic SVM-based stopping rule.

A best example to illustrate the concept of MSFMCS is to use an example of a train formed by a locomotive, cars (or rolling stocks), couplers, and end car, each of which corresponds to the IM, subsequent SFMs, SVM-based stopping rule, and terminated SFM, respectively, as shown in Fig. 3. That is, MSFMCS can be modeled by a train pulling a cascade of train cars with couplers used to connect two cars.

More specifically, IM in MSFMCS initializes the system working as if a locomotive starts its engine. The subsequent SFMs of MSFMCS are simply train cars where an SVM-based stopping rule, denoted by Δ, is attached to the end of each SFM, which corresponds to a train car equipped with a coupler denoted by Δ at its end. The SVM-based stopping rule works as a coupler to connect two train cars and determine whether a new train car is needed to be coupled. The last SFM of MSFMCS is the terminal SFM corresponding to the train end car. Using this train example, each of the above components in MSFMCS can be described in detail as follows.

A. IM

The first component is initialized by IM to serve as an input module. It implements an SC to initialize MSFMCS which may be different from the SVM-based stopping rule used at the end of subsequent SFMs. It is this SC that makes MSFMCS a versatile system because the used SC can be any arbitrary SC. If we interpret the SC used in IM as a locomotive engine, then various SCs correspond to locomotive engines made by different manufacturers. To demonstrate its versatility, three types of heterogeneous SCs, pure-pixel-based SVM, mixed-pixel-based CEM [20]–[22], and feature-extraction-based OTVCA [44], [45] classifiers, will be used for experiments where CEM and OTVCA will have MAP and SVM to perform classification in IM. Since SVM itself is a classifier, it can perform classification in IM. Using the training example, SVM here can be considered as a diesel locomotive which has its own diesel engine built in to provide its own power. In this case, there is no need of stopping rule. In contrast, OTVCA is a feature extraction method and requires a classifier to perform classification, in which case the SVM is used as a stopping rule for IM to perform classification. In both cases, OTVCA and CEM are considered as steam locomotives which need coal carriages to provide their powers with their attached couplers served as stopping rules in IM. It should be noted that since the original CEM in [43] used Otsu's method to perform classification, it resulted in three binarization decisions, no-class assignment, one-class assignment, and multiple-class membership assignment. To resolve this issue, the CEM implemented in this article replaced Otsu's method with MAP to produce a unique class membership assignment for comparison consistency.

As a summary, since the SC being used in the IM varies with different applications, the binarization rules used to produce the classification maps as stopping rules in IM by different SCs are quite different from the SVM-based stopping rules used at the end of subsequent SFMs.

B. SFMs

The second component cascades a number of SFMs corresponding to a cascade of train cars. The SFMs used in MSFMCS are independent modules except that each SFM must be cascaded with its preceding SFM. The SFs used in SFMs can be identical or vary with SFM by SFM. This is like different train cars that can be coupled independently and individually. This advantage provides a significant benefit for hardware design. In addition, the selection of SFs can also be custom-designed such as Gaussian filters, EPF, or Gabor filters.

C. SVM-Based Stopping Rule and Terminal SFM

After the IM, attached to the end of each SFM is an SVM-based stopping rule which corresponds to a coupler equipped at the end of a train car which enables the current train car to

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connect a next train car. It is designed for two goals. One is to allow MSFMCS to keep improving the performance of SC implemented in IM. It provides an automatic rule to determine whether or not the SVM classification results produced by SVM stopping rule can be further improved. In other words, an SVM-based stopping rule sandwiched between two successive SFMs can be interpreted as a railway coupler which connects two train cars where each train car equipped with a coupler corresponds to an SFM with an SVM-based stopping rule attached to the end of the SFM. In this case, the number of train cars to be coupled is determined by the number of SFMs, referred to as the length, d. The other goal is how to determine when the end car will be used to stop coupling, that is, to automatically terminate cascading the system. As a particular case, when an SC performs so well, there is no need of adding a new SFM. Under such a circumstance, the SVM-based stopping rule will immediately terminate the system, in which case MSFMCS is reduced to its IM with no cascaded SFMs. Otherwise, MSFMCS keeps continuing to cascade new SFMs to improve the classification performance until no more improvement can be made in correspondence to no new SFMs needed to be cascaded.

IV. MSFMCS Algorithm

The basic ideas of designing MSFMCS were discussed in detail in Section III. This section details the step-by-step implementation for MSFMCS as follows.

Algorithm 1 MSFMCS Algorithm

1) Initial/input module: $k = 0$.
   a) Randomly select training samples for each of the classes, $\{S_m^{(0)}\}_{m=1}^{M}$, from HSI where $S_m^{(0)}$ is the training sample set of the mth class, C_m.
   b) Implement an SC in the IM to produce spectral classification maps, $\{\text{SCMap}_m\}_{m=1}^{M}$, and also denoted as $\{\text{SVMap}_m^{(0)}\}_{m=1}^{M} = \{\text{SCMap}_m\}_{m=1}^{M}$.
2) Let $k \leftarrow k + 1$. At the kth SFM, apply a set of M spatial filters $\{\text{SF}_m^{(k)}\}_{m=1}^{M}$ to $\{\text{SVMap}_m^{(k-1)}\}_{m=1}^{M}$ to produce $\{\text{SFMap}_m^{(k)}\}_{m=1}^{M}$.
3) Implement SVM on $\{\text{SFMap}_m^{(k)}\}_{m=1}^{M}$ using training samples for each of the classes, $\{S_m^{(k)}\}_{m=1}^{M}$, generated by either FTS or RTS to produce M SVM classification maps, $\{\text{SVMap}_m^{(k)}\}_{m=1}^{M}$. If FTS is used, then $S_m^{(k)} = S_m^{(0)}$ for all k. If RTS is used, then to produce M SVM classification maps, $\{S_m^{(k)}\}_{m=1}^{M} \neq \{S_m^{(k-1)}\}_{m=1}^{M}$ for $k \geq 1$.
4) Check if

$$\text{OA}(\text{SVMap}_m^{(k)}) \leq \text{OA}(\text{SVMap}_m^{(k-1)}) \tag{2}$$

If it is, the algorithm is terminated. Output the final classification maps $\{\text{SVMap}_m^{(k-1)}\}_{m=1}^{M}$ and the length d is $(k-1)$. Otherwise, go to step 2.

According to the above algorithm, the IM in step 1 performs any selected SC to produce the classification maps $\{\text{SCMap}_m\}_{m=1}^{M}$. This means that any SC can be used as an initial SC for MSFMCS without using SFs. Step 2 performs spatial filtering on $\{\text{SVMap}_m^{(k-1)}\}_{m=1}^{M}$ to produce $\{\text{SFMap}_m^{(k)}\}_{m=1}^{M}$ which will be an input to the SVM-based stopping rule described in step 3. Step 4 is a decision made on the results obtained by step 3 to determine whether or not the cascading process should be continued or terminated. This step also determines the length of MSFMCS, d.

V. Allocation of Training Sample Class Size

A commonly used approach for selection of training samples is RTS which randomly selects the training samples according to a certain percentage from each class. However, there are two major problems associated with this approach. One is the determination of how much percentage needs to be used for training sample selection. The other is the training class size determined by the same chosen percentage, which may result in insufficient training samples to be selected for small classes. While the first issue is generally determined empirically, the second issue is closely related to imbalanced class and can be addressed by recent works developed in [13], [46], and [47] on how to select individual size of training samples for each class, namely, n_m^{training} for class C_m. The approach presented in this section is primarily developed by Kang [48], referred to as Kang's method (KM) which turns out to be an effective means of allocating n_m^{training} provided that the total number of training data samples, N^{training}, is given as *a priori*. The details of its step-by-step implementation can be briefly described below.

Algorithm 2 *Kang's Method (KM)*

1) Initial condition: for a given N^{training} obtained from a given percentile (%), determine $n_{\text{average}} = \lfloor N^{\text{training}}/M \rfloor$.
2) If $\lfloor n_m/2 \rfloor > n_{\text{average}}$ where n_m is the number of data samples in class C_m, then $\bar{n}_m^{\text{training}} = n_{\text{average}}$. In this case, n_m^{training} is over-sampling for large classes. Otherwise, it is under-sampling. We let $\bar{n}_m^{\text{training}} = \lfloor n_m/2 \rfloor$ for small classes.
3) If $\bar{N}^{\text{training}} = \sum_{m=1}^{M} \bar{n}_m^{\text{training}} < N^{\text{training}}$, then let $\tilde{N}^{\text{training}} = N^{\text{training}} - \bar{N}^{\text{training}}$.
4) Randomly select $\tilde{N}^{\text{training}}$ training samples among the remaining samples of M classes obtaining $\tilde{n}_m^{\text{training}}$. Then, the final training sample set for class C_m is

$$n_m^{\text{training}} = \bar{n}_m^{\text{training}} + \tilde{n}_m^{\text{training}} \tag{3}$$

In addition to KM, there are also other methods developed in [47] using the concept of class significance that can also be used for the same purpose. Since these methods are more involved, KM was chosen to allocate n_m^{training} for simplicity.

VI. Images and Classification Measures to Be Used for Experiments

Although there are many image scenes available on the website [49] that can be used for HSIC, three real

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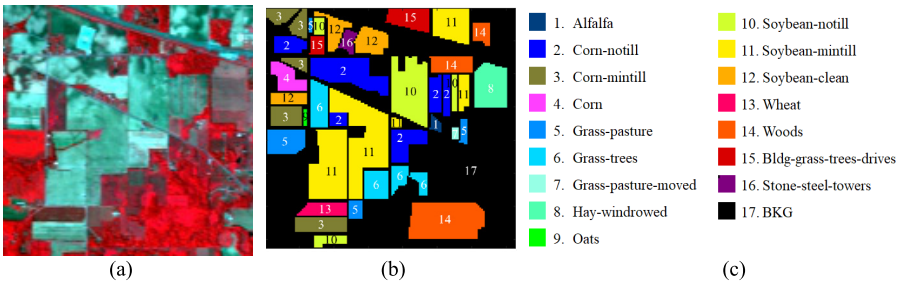


Fig. 4. Purdue Indiana Indian Pines. (a) False-color composite image. (b) Reference map. (c) Classes by colors.

hyperspectral images are of particular interest. One is Purdue's Indiana Indian Pines which has an imbalanced class issue. A second one is University of Pavia which has a very complex BKG. A third and the last one is Houston data, which is currently available on GRSS-DASE. These images were used to conduct a comprehensive study and analysis.

A. Purdue's Indiana Indian Pines

The first image used for experiment is Purdue University's Indiana Indian Pines shown in Fig. 4(a). It is a well-known AVIRIS image scene with the ground truth of 16 class maps plus BKG in Fig. 4(b) and class labels in Fig. 4(c). It was recorded in June 1992 using 200 bands with 20 water absorption bands (bands 104–108 and 150–163, 220) removed. Table I tabulates all the 16 classes of interest along with the number of data samples in each class where four small classes, classes 9, 7, 1, and 16, contain less than 100 data samples and three large classes, classes 14, 2, and 11, contain more than 1000 data samples. These seven classes can be used to demonstrate the issue of imbalanced classes in classification performance.

B. University of Pavia

A second hyperspectral image used for experiments is shown in Fig. 5(a) which is different from the Purdue data considered in Section VI-A. It was collected by the ROSIS-03 satellite sensor over an urban area surrounding the University of Pavia, Italy. It is of size 610 × 340 × 115 with a spatial resolution of 1.3 m per pixel and a spectral coverage ranging from 0.43 to 0.86 μm with spectral resolution of 4 nm (12 most noisy channels were removed before experiments). Nine classes of interest are considered for this image. Fig. 5(b) provides its ground-truth map along with color class labels in Fig. 5(c). According to [14], the overall PR defined in Section VI-C) from this data set was very low, less than 30% twice lower than that obtained from the Purdue data set indicating that this scene has more complicated BKG than the other. Table II also tabulates the number of data samples in parentheses collected for each class where most classes are relatively very large except for class 9 with 947 < 1000 data samples. So there is no imbalanced class issue.

C. Houston Data

A third image scene is shown in Fig. 6 was collected over the campus area of University of Houston, Houston, TX,

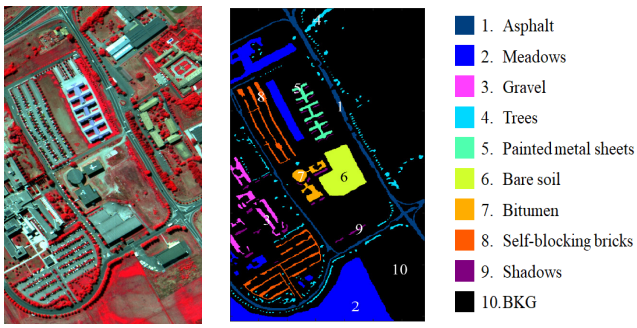


Fig. 5. University of Pavia. (a) False-color composite image. (b) Reference map. (c) Classes by colors.

USA, and its neighboring urban area as shown in Fig. 6. The image was captured by compact airborne spectrographic image (CASI) sensor in 2012. The data set was published in 2013 IEEE GRSS Data Fusion Contest. It has a size of 349 × 1905 × 144 with 144 spectral bands ranging 380–1050 nm and a spatial resolution of 2.5 m. Table III tabulates the number of data samples in parentheses collected for each class.

Table IV tabulates the number of training samples, n_{train}, and the number of test samples, n_{test}, used for experiments conducted for Purdue's data, U. of Pavia, and Houston data in this section.

D. Classification Measures to Be Used for Experiments

The following several classification measures defined in [14] will be used for experiments. Let

M = number of classes.

C_m = set of data samples in the mth class, by ground truth.

$\hat{C}_j$ = set of data samples classified into the jth class, by classified map.

$n_m = \sum_{j=1}^{M} n_{jm}$ = number of data samples in C_m.

n_{jm} = number of data samples in the mth class, C_m, but actually classified into the jth class, $\hat{C}_j$.

N = total number of data samples, $N = \sum_{m=1}^{M} n_m$.

n_{mm} = number of signal samples in the mth class, C_m, correctly classified into the mth class, $\hat{C}_m$.

In traditional HSIC, the popular performance measurements are class accuracy (A) and OA, given by

$$P_{\text{A}}(C_m) = \text{accuracy of the } m\text{th class} = \frac{n_{mm}}{\sum_{j=1}^{M} n_{jm}} = \frac{n_{mm}}{n_m} \tag{4}$$

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TABLE I
CLASS LABELS OF INDIAN PINES WITH THE NUMBER OF DATA SAMPLES IN EACH CLASS

class 1 (46)	Alfalfa	class 7 (28)	Grass/pasture-mowed	class 13 (205)	Wheat
class 2 (1428)	Corn-notill	class 8 (478)	Hay-windrowed	class 14 (1265)	Woods
class 3 (830)	Corn-mintill	class 9 (20)	Oats	class 15 (386)	Bldg-grass green-drives
class 4 (237)	Corn	class 10 (972)	Soybeans-notill	class 16 (93)	Stone-steel towers
class 5 (483)	Grass/pasture	class 11 (2455)	Soybeans-mintill	class 17 (10776)	BKG
class 6 (730)	Grass/trees	class 12 (593)	Soybeans-clean		

TABLE II
CLASS LABELS OF UNIVERSITY OF PAVIA WITH THE NUMBER OF DATA SAMPLES IN EACH CLASS

class 1 (6631)	Asphalt	class 5 (1345)	Painted metal sheets	Class 9 (947)	Shadows
class 2 (18649)	Meadows	class 6 (5029)	Bare soil	Class 10 (164624)	BKG
class 3 (2099)	Gravel	class 7 (1330)	Bitumen		
class 4 (3064)	Trees	class 8 (3682)	Self-blocking bricks		

TABLE III
CLASS LABELS OF HOUSTON DATA WITH THE NUMBER OF DATA SAMPLES IN EACH CLASS

class 1 (1251)	Healthy grass	class 7 (1268)	Residential	class 13 (469)	Parking lot 2
class 2 (1254)	Stressed grass	class 8 (1244)	Commercial	class 14 (428)	Tennis court
class 3 (697)	Synthetic grass	class 9 (1252)	Road	class 15 (660)	Running track
class 4 (1244)	Trees	class 10 (1227)	Highway	class 16 (649816)	BKG
class 5 (1242)	Soil	class 11 (1235)	Railway		
class 6 (325)	Water	class 12 (1233)	Parking lot 1		

$$P_{\mathrm{OA}} = \frac{1}{N}\sum_{m=1}^{M} n_{mm} = \sum_{m=1}^{M} \frac{n_m}{N} P_{\mathrm{A}}(C_m) \tag{5}$$

which shows that P_{OA} uses sample ratios as weights to average the accuracy of each class. In addition, we can consider all classes to be equally likely by letting $(n_m/N) = (1/M)$ to further define P_{AA} as AA by the number of classes, M, expressed as

$$P_{\mathrm{AA}} = \frac{1}{M}\sum_{m=1}^{M} \frac{n_{mm}}{n_m} = \frac{1}{M}\sum_{m=1}^{M} P_{\mathrm{A}}(C_m). \tag{6}$$

Another measure is PR given by

$$P_{\mathrm{PR}}(C_m) = \text{precision rate of } C_m = \frac{\hat{n}_{mm}}{\hat{n}_m} \tag{7}$$

where $\hat{n}_m = \sum_{j=1}^{M} \hat{n}_{mj}$ is the total number of data samples that be classified into the mth class and $\hat{n}_{mj}$ is the number of data samples classified into the mth class but supposed to be in the jth class. So the average precision (APR), P_{APR}, in correspondence to P_{AA} in (6) is defined by

$$P_{\mathrm{APR}} = \frac{1}{M}\sum_{m=1}^{M} P_{\mathrm{PR}}(\hat{C}_m). \tag{8}$$

Since n_{mm} is identical to $\hat{n}_{mm}$ and $\hat{N} = \sum_{i=1}^{M} \hat{n}_i$ is equal to $N = \sum_{i=1}^{M} n_i$, an overall precision (OPR), P_{OPR} can also be defined as the counterpart of P_{OA} in (5) by

$$P_{\mathrm{OPR}} = \sum_{i=1}^{M} \left(\frac{\hat{n}_i}{\hat{N}}\right) P_{\mathrm{PR}}(\hat{C}_i) = \sum_{i=1}^{M} \left(\frac{n_i}{N}\right) P_{\mathrm{A}}(C_i) = P_{\mathrm{OA}} \tag{9}$$

TABLE IV
NUMBERS OF TRAINING SAMPLES, n_{train}, AND NUMBERS OF TEST SAMPLES, n_{test}, USED FOR PURDUE DATA, UNIVERSITY OF PAVIA, AND HOUSTON DATA

Class	Purdue		University of Pavia		Houston	
	n_{train}	n_{test}	n_{train}	n_{test}	n_{train}	n_{test}
1	23	23	48	6583	50	1201
2	44	1384	50	18599	51	1203
3	36	794	47	2052	50	647
4	32	205	47	3017	50	1194
5	34	449	47	1298	50	1192
6	34	696	47	4982	50	275
7	14	14	47	1283	50	1218
8	34	444	48	3634	50	1194
9	10	10	47	900	50	1202
10	36	936			50	1177
11	44	2411			50	1185
12	35	558			50	1183
13	32	173		-	50	419
14	36	1229			50	378
15	34	352			50	610
16	34	59				-
Total	512	9737	428	42348	751	14278

which implies that P_{OPR} is essentially the same P_{OA} provided that BKG class samples are not included for classification.

It should be noted that PR, also known as the user's accuracy, is proposed to address BKG issue, while many existing classification methods have removed all unlabeled data samples as BKG from classification. PR is the one that can include all data samples for evaluation.

VII. EXPERIMENTAL RESULTS AND DISCUSSION

To validate the utility of MSFMCS, various scenarios were performed for experiments and evaluated for a comparative

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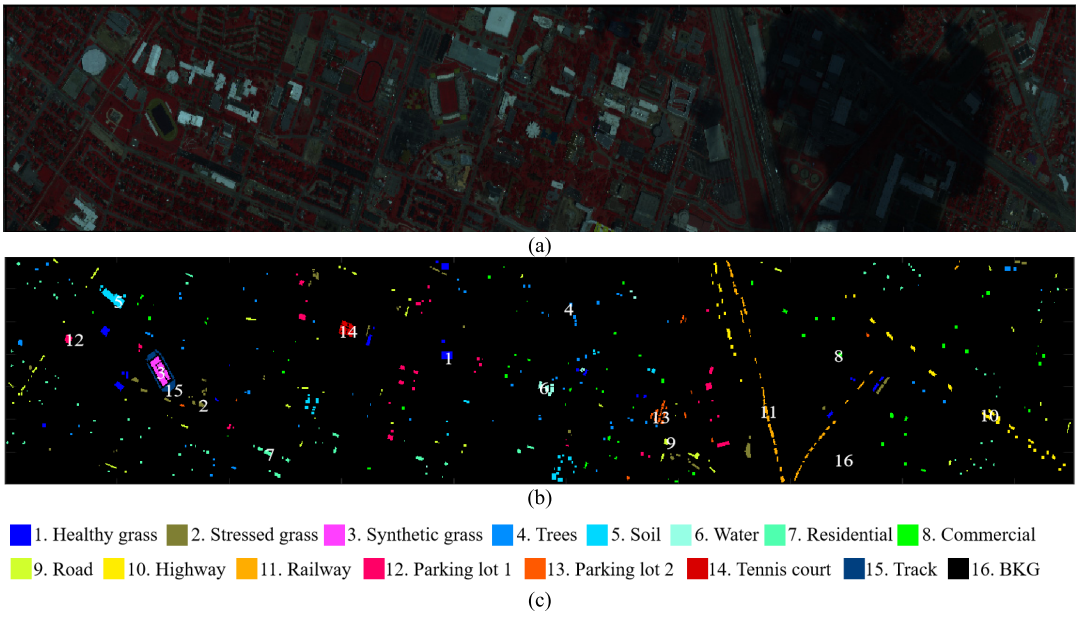


Fig. 6. Houston data. (a) False-color composite image. (b) Reference map. (c) Classes by colors.

study and analysis. Since EPF-G-g has been used throughout experiments, it will be abbreviated as EPF for simplicity.

1) *Versatility of SC Used in IM*: Three different types of heterogeneous classifiers were used as SCs in IM:
   a) pixel-based classifier: SVM;
   b) mixed-pixel classifier: CEM with MAP to produce classification maps;
   c) feature-based classifiers: Orthogonal total variation component analysis (OTVCA) [44], [45] with SVM used as a classifier.

   Since CEM is linear and SVM is a nonlinear kernel-based classifier, SVM would do better than CEM. On the other hand, OTVCA is a feature-extraction-based SVM classifier, and it would certainly perform better than SVM. The experiments conducted in Section VIII confirm these facts as expected.
2) *First SFM*: Two versions of the first SFM can be implemented for MSFMCS.
   a) MAP used to terminate MSFMCS at the first SFM:
      i) SVM-EPF-MAP in [7] (EPF-G-g).
      It should be noted that there are four EPF-based methods introduced in [7], EPF-B-c, EPF-G-c, EPF-B-g, and EPF-G-g were proposed with "B" and "G" used to specify the bilateral filter and guided filter, respectively, and "g" and "c" indicate that PC_1 and color composite of three PCs are used as reference images. In this article, only EPF-G-g was used for experiments and the other three EPF-based methods can be used as well for this purpose.
      ii) CEM-Gaussian-MAP;
      iii) OTVCA-SVM-EPF-MAP (OTVCA-EPF-G-g).
   b) SVM used as a stopping rule to continue MSFMCS to next cascaded SFM:
      i) SVM-EPF-G-g-SVM (SVM-EPF-SVM);
      ii) CEM-Gaussian-SVM;
      iii) OTVCA-SVM-EPF-SVM.
3) *SFs*: When MSFMCS was compared with EPF, the SFs implemented in MSFMCS were also EPF for comparison.
4) *Training Sample Selection*: FTS/RTS.

Because this SVM-based stopping rule also requires a set of training sample set to perform classification, there are two training sampling strategies proposed in [14] that can be used in MSFMCS to select training samples. One is the FTS strategy which randomly selects training samples in the first SFM and then uses the same training sample set for the SVM-based stopping rule used in the remaining SFMs. The other is the RTS strategy which randomly selects a training sample set for the SVM-based stopping rule used by each new SFM in MSFMCS. In other words, the training sample set used in each SFM is completely random and independent of the training samples selected in the previous SFMs. As a result, the length defined as the number of SFMs resulting from FTS and RTS is different. So in the following experiments, the average length (AL) was documented for comparison where AL was calculated by running MSFMCS using FTS and RTS five times where the training samples were randomly selected according to allocation of the training sample size for each class determined by KM described in Section IV for comparison.

A. Purdue Indiana Indian Pines

Fig. 7(a)–(e) shows the classification maps produced for the Purdue data by MSFMCS using SVM as the SC in the IM where MSFMCS was run five times and the results were the best cases among the five different runs with OA. Specifically, Fig. 7(a)–(c) shows the classification maps obtained by SVM alone without using SFs, the results obtained by EPF-G-g

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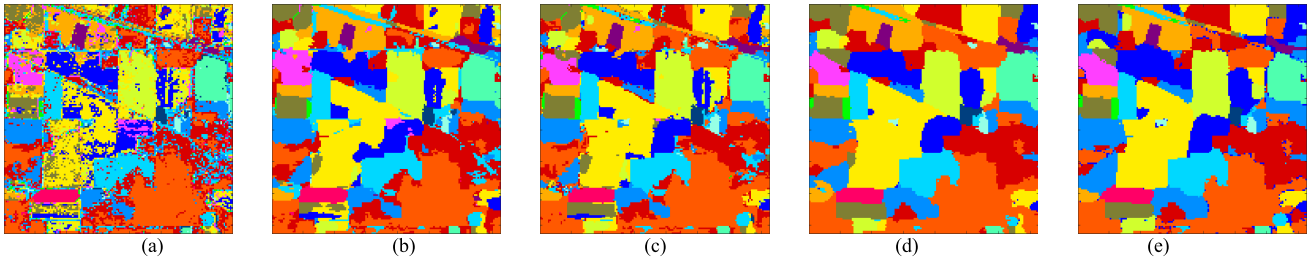


Fig. 7. Classification maps with OA for the Purdue data produced by MSFMCS using SVM as the SC in the IM. (a) SVM = 73.38%. (b) EPF-G-g = 86.55%. (c) First SFM = 91.54%. (d) Terminal SFM (FTS) = 94.90%. (e) Terminal SFM (RTS) = 96.71%.

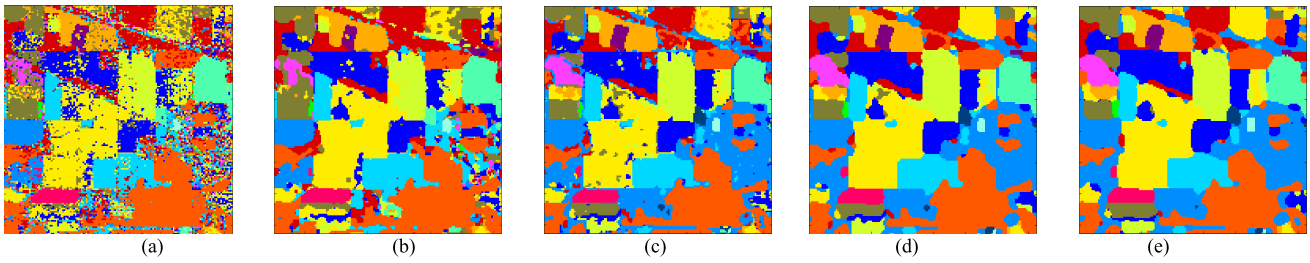


Fig. 8. Classification maps with OA for the Purdue data produced by MSFMCS using CEM as the SC in the IM. (a) CEM-MAP = 77.48%. (b) CEM-Gaussian-MAP = 85.26%. (c) First SFM = 92.06%. (d) Terminal SFM (FTS) = 95.56%. (e) Terminal SFM (RTS) = 97.05%.

with MAP used as the stopping rule, and the results obtained using SVM as the stopping rule in the first SFM, respectively. As we can see, using SVM alone without SFs performs the worst with only 73.38% of OA followed by EPF-G-g with 86.55% of OA, while the best one was produced by the first SFM using SVM as the stopping rule with 91.54% of OA. Fig. 7(d) and (e) further shows the classification maps obtained by MSFMCS with the terminal SFMs using FTS and RTS, respectively. It clearly showed that after MSFMCS was terminated, OA was further improved to 94.90% by FTS and 96.71% by RTS. In addition, MSFMCS also showed its advantage of using its first SFM over EPF-G-g used in the first SFM with improvement by nearly 5%. Considering the fact that only 5% of the data samples were used as the training samples, the results in Fig. 7(d) and (e) were significant. For more detailed comparisons, Table V tabulates P_{A}, P_{PR}, P_{AA}, P_{OA}, P_{APR}, P_{OPR}, computing times, and AL of running MSFMCS five times using SVM in the IM, first SFM, and terminal SFM with 5% training samples using KM for the Purdue data where the best results are boldfaced. Like P_{OA}, P_{AA} was also significantly improved. Interestingly, PR was not improved as much due to BKG effects. As for computing time, MSFMCS only used less twice time than one single module to terminate its process.

Similarly, Figs. 8(a)–(e) and 9(a)–(e) show the classification maps produced for the Purdue data by MSFMCS using CEM-MAP and OTVCA-SVM as the SC in the IM, respectively. Tables VI and VII also tabulate their P_{A}, P_{PR}, P_{AA}, P_{OA}, P_{APR}, P_{OPR}, computing times, and AL of running MSFMCS five times using CEM-MAP and OTVCA-SVM as the SC in the IM, first SFM, and terminal SFM with 5% training samples using KM for the Purdue data where the best results are boldfaced. The same conclusion drawn from SVM in Fig. 7(a)–(e) and Table V could also be applied to CEM-MAP and OTVCA-SVM. There is an interesting observation. When OTVCA-SVM was used as the SC in the IM, the improvement by MSFMCS was not as significant as that using SVM and CEM-MAP as the SC in the IM. In particular, OTVCA did not take advantage of MSFMCS's first SFM to improve the OA performance. This was mainly due to the fact that OTVCA was used as a feature extraction method prior to SVM. As a result, MSFMCS could still improve OTVCA-SVM but not as much as it did for SVM and CEM-MAP. These experiments further demonstrated that MSFMCS can improve any arbitrary SC. However, the level of its improvement is really determined by the SC used in the IM. If SC is very effective such as OTVCA-SVM, the room of improvement will be limited. On the other hand, if SC is not good, the improvement resulting from the use of MSFMCS will be great and tremendous such as SVM and CEM-MAP. As a result, it took MSFMCS more SFMs for SVM and CEM-MAP than that for OTVCA to terminate the cascaded system. All these experiments made a very good sense because CEM is a linear classifier compared with SVM which is a nonlinear kernel-based classifier. Of course, SVM did better than CEM in classification. Furthermore, OTVCA is a feature extraction method coupled with SVM. Accordingly, OTVCA-SVM certainly performed better than SVM alone.

B. University of Pavia

The same experiments conducted for the Purdue data were also performed for U. of Pavia. Figs. 10(a)–(e)–12(a)–(e) show the classification maps produced for U. of Pavia by MSFMCS using SVM, CEM-MAP, and OTVCA-SVM as the SC in the IM, respectively, where MSFMCS was run five times and the results were the best cases among the five different runs with OA. Tables VIII–X also tabulate P_{A}, P_{PR}, P_{AA}, P_{OA}, P_{APR},

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TABLE V

P_A, P_{PR}, P_{AA}, P_{OA}, P_{APR}, P_{OPR}, COMPUTING TIMES, AND AL OF MSFMCS USING SVM WITH FIVE RUNS IN THE IM, FIRST SFM, AND TERMINAL SFM WITH 5% TRAINING SAMPLES USING KM FOR THE PURDUE DATA

Class	IM SVM		1st SFM EPF-G-g		1st SFM MSFMCS		Terminal SFM (FTS) MSFMCS AL = 6.40±2.07 modules		Terminal SFM (RTS) MSFMCS AL = 5.60±1.82 modules	
	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$
1	97.39±0.97	24.03±3.76	100±0	39.79±5.02	98.69±1.19	44.32±10.5	100±0	37.28±9.18	99.56±0.97	39.62±11.2
2	66.65±3.97	51.45±3.68	80.89±4.77	57.41±5.97	86.48±5.26	59.44±3.58	88.61±5.69	57.83±6.11	93.01±3.95	56.07±3.96
3	62.04±2.89	46.71±4.41	70.12±6.20	64.00±4.81	87.25±5.13	62.15±4.52	94.98±5.06	68.49±7.14	95.42±4.82	69.21±4.93
4	83.45±8.35	25.14±6.95	100±0	41.54±10.3	98.56±1.53	61.38±10.7	99.66±0.55	61.43±10.2	99.83±0.23	61.84±6.43
5	86.16±1.60	26.76±3.72	90.72±1.22	36.09±7.20	94.07±2.27	29.39±3.04	95.44±3.00	28.40±9.54	97.10±2.37	27.00±5.68
6	91.89±0.72	32.82±4.31	99.47±0.87	41.56±7.45	99.20±0.51	53.68±6.84	99.47±0.58	59.60±5.10	99.83±0.11	58.18±4.17
7	97.14±2.98	17.88±5.55	96.42±1.24	49.22±15.9	98.57±1.95	39.62±23.7	98.57±1.95	44.78±15.8	97.85±1.95	43.14±12.8
8	96.06±2.32	78.23±4.51	99.83±0.27	71.69±2.36	99.91±0.18	72.55±3.26	100±0	65.35±5.16	99.87±0.28	66.49±3.52
9	96±5.47	9.624±3.16	100±0	39.02±10.6	100±0	23.17±12.2	100±0	21.96±8.89	100±0	20.47±4.54
10	73.10±5.80	50.81±4.05	84.81±5.79	62.93±6.83	90.39±4.93	64.74±7.58	90.32±4.93	64.51±7.25	90.67±4.59	64.91±4.18
11	63.14±3.93	67.92±6.90	82.61±3.72	76.17±6.30	87.13±3.27	77.26±5.81	93.31±2.62	77.25±5.08	93.90±2.02	78.22±3.12
12	71.12±5.92	36.66±4.66	93.59±4.86	47.17±5.18	95.07±1.17	54.70±7.12	98.17±0.81	57.88±9.31	98.88±0.43	59.61±7.12
13	98.04±1.14	39.42±7.29	99.41±0.21	71.63±3.36	99.41±0.21	74.81±6.95	99.31±0.26	70.24±5.29	99.41±0.21	71.65±4.28
14	88.82±3.36	34.68±0.95	94.98±2.96	33.61±1.97	97.29±0.97	32.02±3.14	99.81±0.16	32.36±3.82	99.73±0.24	31.49±1.90
15	63.57±2.96	7.223±0.40	86.37±7.14	8.166±0.59	92.74±5.53	9.546±1.30	98.44±1.82	11.83±2.21	99.01±0.64	12.22±1.23
16	97.41±2.22	26.20±5.05	99.78±0.48	26.71±5.27	99.56±0.58	30.38±6.65	100±0	32.66±10.3	99.56±0.58	34.33±6.11
P_{AA}	83.25±0.85		92.44±0.66		95.27±0.58		97.26±0.33		**97.73±0.48**	
P_{OA}	74.59±1.05		87.30±1.15		91.80±1.13		95.02±0.49		**95.99±0.94**	
P_{APR}	35.97±0.58		47.92±1.55		49.32±3.00		49.49±1.77		**49.65±0.68**	
P_{OPR}	36.36±0.51		42.55±0.56		44.75±0.55		46.32±0.24		**46.79±0.45**	
Time	33.48±1.89		33.58±1.89		38.98±2.13		64.14±10.9		59.66±8.28	

TABLE VI

P_A, P_{PR}, P_{AA}, P_{OA}, P_{APR}, P_{OPR}, COMPUTING TIMES, AND AL OF MSFMCS USING CEM WITH FIVE RUNS IN THE IM, FIRST SFM, AND TERMINAL SFM WITH 5% TRAINING SAMPLES USING KM FOR PURDUE DATA

Class	IM CEM-MAP		1st SFM CEM-Gaussian-MAP		1st SFM MSFMCS		Terminal SFM(FTS) MSFMCS AL = 14.60±2.07 modules		Terminal SFM(RTS) MSFMCS AL = 7.60±2.79 modules	
	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$
1	4.35	50.00	0.00	0.00	99.56±0.97	44.28±9.70	99.56±0.97	47.96±8.05	99.56±0.97	46.05±4.88
2	76.54	43.91	89.71	52.16	89.98±3.55	66.22±7.63	93.17±3.06	68.66±6.79	94.39±2.72	65.34±5.06
3	70.36	29.23	76.39	34.55	85.51±1.83	64.53±4.15	90.09±1.59	69.11±2.72	93.71±1.95	65.82±6.32
4	50.21	24.04	57.81	51.70	98.73±0.66	67.07±9.55	99.57±0.29	65.18±8.99	99.57±0.51	70.20±3.36
5	68.12	34.20	65.84	36.55	97.92±1.92	10.10±0.84	98.46±2.00	10.33±1.07	98.46±1.50	11.42±0.98
6	95.07	34.67	99.59	33.03	99.28±0.32	60.37±4.38	99.72±0.21	60.76±2.54	99.89±0.15	59.01±4.34
7	89.29	30.49	100.00	52.83	100±0	46.97±8.90	100±0	46.22±9.09	100±0	45.69±6.19
8	100.00	61.20	100.00	55.97	99.95±0.09	69.37±5.25	100±0	70.96±4.29	100±0	70.95±1.25
9	90.00	13.74	75.00	35.71	100±0	56.97±11.20	100±0	52.07±4.20	99±2.23	49.98±5.32
10	67.49	40.10	79.94	53.85	91.17±2.32	59.87±5.59	93.74±2.10	64.21±5.79	95.59±2.69	61.55±4.27
11	73.73	56.05	85.01	62.71	82.87±6.07	71.19±4.91	90.89±4.00	72.77±3.47	92.72±2.15	70.66±5.21
12	70.32	46.59	76.56	56.96	92.44±3.61	57.76±6.04	94.16±4.05	59.31±6.28	94.83±3.79	60.39±6.18
13	99.51	56.82	99.51	68.23	99.80±0.26	67.31±12.10	99.90±0.21	63.36±5.74	100±0	66.62±4.13
14	86.56	31.48	89.72	29.55	96.83±0.83	35.10±3.44	98.37±1.06	35.26±3.75	98.07±0.99	35.09±2.52
15	88.86	14.90	98.96	14.92	94.61±2.29	24.78±1.75	98.29±0.92	27.36±3.84	99.48±0.89	27.34±2.70
16	79.57	40.66	87.10	44.75	99.78±0.48	57.84±6.18	99.78±0.48	60.34±4.17	99.78±0.48	59.48±4.08
P_{AA}	75.62		80.07		95.53±0.59		97.23±0.41		**97.81±0.53**	
P_{OA}	77.48		85.26		91.27±1.69		94.76±1.29		**95.89±0.88**	
P_{APR}	38.01		42.72		53.73±2.23		**54.62±1.67**		54.10±0.58	
P_{OPR}	37.77		41.56		44.49±0.82		46.19±0.63		**46.74±0.43**	
Time	1.34		1.39		6.45±0.15		55.90±7.12		30.77±10.50	

P_{OPR}, computing times, and AL of running MSFMCS five times using SVM, CEM-MAP, and OTVCA-SVM as the SC in the IM, first SFM, and terminal SFM with 1% training samples using KM for U. of Pavia where the best results are boldfaced. Analogous to the Purdue data experiments, MSFMCS could improve OA of SVM and CEM-MAP significantly by simply using one single SFM by nearly 10% and only 1%–3% further improvement after more subsequent SFMs were used until MSFMCS was terminated. However, CEM-MAP did take advantage of MSFMCS's first SFM to improve its performance significantly. For example, it improved its OA over that using CEM in its first SFM by 10% as shown in Table IX compared with no improvement using EPF-G-g used in the first SFM in Table VIII. Nevertheless, MSFMCS eventually managed to improve EPF-G-g's OA by nearly 2% at the end in Table VIII.

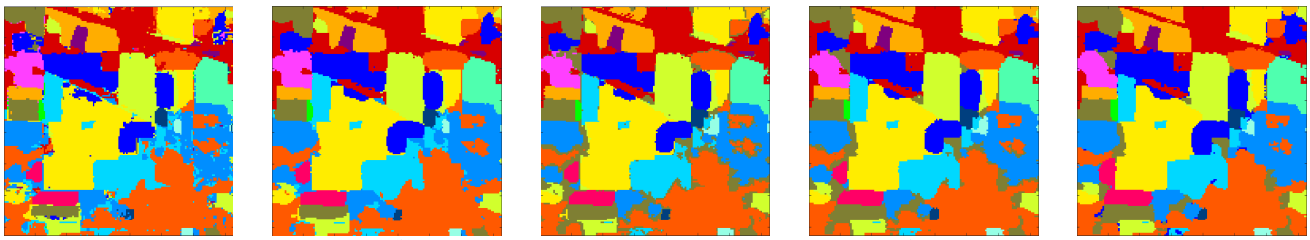
With using OTVCA as the SC in the IM, MSFMCS could also improve its OA by 3%–4% simply taking advantage of its first SFM, but the further improvement was very limited afterward. As for PR, unlike the Purdue data experiments, very little improvement could be gained by MSFMCS due to the fact that U. of Pavia has more complicated BKG than the

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TABLE VII

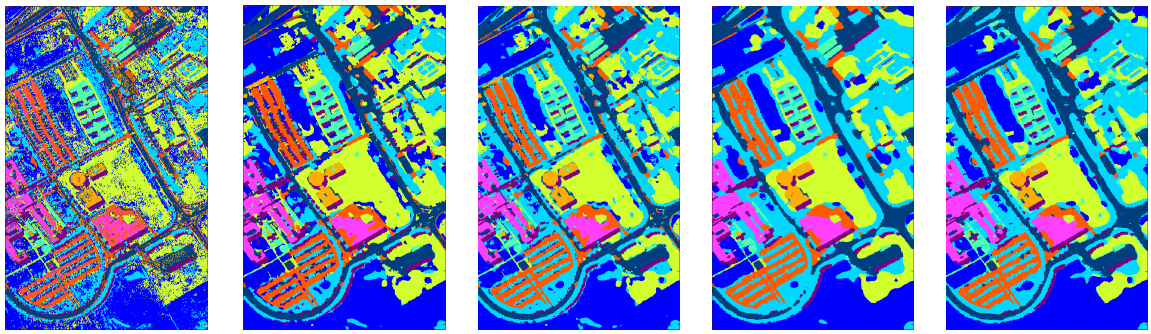
P_A, P_{PR}, P_{AA}, P_{OA}, P_{APR}, P_{OPR}, COMPUTING TIMES, AND AL OF MSFMCS USING OTVCA WITH FIVE RUNS IN THE IM, FIRST SFM, AND TERMINAL SFM WITH 5% TRAINING SAMPLES USING KM FOR PURDUE DATA

Class	IM OTVCA-SVM		1st SFM OTVCA-EPF-G-g		1st SFM MSFMCS		Terminal SFM(FTS) MSFMCS AL = 2.60±0.89 modules		Terminal SFM(RTS) MSFMCS AL = 3.40±1.67 modules	
	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$
1	98.26±1.81	40.89±17.5	100±0	40.97±11.5	100±0	38.46±11.0	100±0	36.17±8.03	100±0	36.66±5.66
2	90.26±5.19	66.40±8.02	92.28±4.49	66.84±7.43	92.50±4.28	66.81±8.24	93.09±3.58	64.88±7.60	93.34±4.60	64.86±7.06
3	89.66±5.81	65.59±7.61	90.86±4.71	65.36±7.18	95.46±4.40	54.57±15.7	95.44±4.19	56.42±12.4	98.02±1.14	58.92±7.86
4	97.89±2.92	52.41±8.28	100±0	55.81±9.20	98.39±2.89	61.92±9.67	99.66±0.55	60.94±8.74	99.32±1.10	64.24±4.32
5	93.04±8.46	17.41±1.83	93.62±8.84	17.33±1.71	93.54±8.43	17.58±3.95	93.91±7.10	16.97±3.69	98.09±1.85	16.47±2.00
6	99.75±0.34	41.21±4.11	100±0	45.78±4.20	99.94±0.12	48.40±6.36	99.97±0.06	50.47±6.98	99.97±0.06	51.46±8.61
7	100±0	33.37±15.0	96.42±1.24	39.99±14.5	97.85±1.95	30.94±12.5	97.85±1.95	29.59±12.5	97.85±1.95	23.68±3.14
8	99.91±0.11	76.25±1.73	100±0	75.49±0.67	100±0	73.27±1.04	100±0	71.81±0.58	100±0	70.94±2.03
9	100±0	24.05±12.3	100±0	37.06±7.88	100±0	32.38±5.74	100±0	34.40±6.10	100±0	29.39±2.31
10	95.08±0.68	57.37±5.10	95.39±1.62	62.89±6.60	96.15±1.51	62.78±9.04	94.38±2.91	62.56±7.97	94.77±2.38	64.73±8.82
11	89.58±5.03	79.07±4.25	92.83±4.82	79.85±3.23	92.59±4.61	79.69±3.05	93.89±4.99	78.89±2.24	94.74±4.35	77.33±2.05
12	96.72±1.32	57.98±6.58	99.15±0.57	60.40±8.22	99.19±0.56	58.53±6.91	99.39±0.40	57.03±6.71	99.19±0.38	58.62±7.10
13	99.51±0	58.79±11.1	99.41±0.21	58.45±11.8	99.51±0.34	56.88±11.2	99.51±0.34	54.38±12.5	99.60±0.21	50.63±13.1
14	95.38±3.39	29.50±1.81	96.45±4.30	28.55±1.77	96.30±3.95	31.11±3.05	96.96±4.00	30.99±2.65	99.30±0.56	32.07±1.38
15	94.87±1.88	17.94±2.80	99.58±0.65	17.70±2.14	99.32±0.43	19.15±2.61	99.53±0.53	20.38±2.24	99.68±0.42	21.16±3.25
16	99.56±0.96	48.96±7.49	99.78±0.48	61.05±4.85	99.78±0.48	59.21±8.84	99.78±0.48	60.84±10.9	99.56±0.58	62.87±11.9
P_{AA}	96.22±1.04		97.23±0.88		97.53±1.00		97.71±1.01		**98.34±0.24**	
P_{OA}	93.47±1.79		95.20±1.53		95.55±1.66		95.93±1.65		**96.88±0.84**	
P_{APR}	47.95±2.39		**50.84±2.25**		49.48±1.82		49.17±1.56		49.00±1.73	
P_{OPR}	45.56±0.87		46.41±0.74		46.58±0.81		46.76±0.80		**47.23±0.41**	
Time	9.19±0.48		9.46±0.70		15.86±1.58		28.10±6.88		29.99±10.12	



(a) (b) (c) (d) (e)

Fig. 9. Classification maps with OA for the Purdue data produced by MSFMCS using OTVCA as the SC in the IM. (a) OTVCA-SVM = 94.30%. (b) OTVCA-EPF-G-g = 95.69%. (c) First SFM = 96.25%. (d) Terminal SFM (FTS) = 96.77%. (e) Terminal SFM (RTS) = 97.48%.



(a) (b) (c) (d) (e)

Fig. 10. Classification maps with OA for University of Pavia produced by MSFMCS using SVM as the SC in the IM. (a) SVM = 84.83%. (b) EPF-G-g = 95.65%. (c) First SFM = 95.48%. (d) Terminal SFM (FTS) = 98.58%. (e) Terminal SFM (RTS) = 98.16%.

Purdue data do. These experiments also demonstrated that the effectiveness of SC used in the IM had tremendous impact on the improvement by MSFMC. That is, the less effective an SC is, the greater improvement MSFMCS can provide.

C. Houston Data

The same experiments conducted for the Purdue data and U. of Pavia were also performed for the Houston data. Figs. 13(a)–(e)–15(a)–(e) show the classification maps produced for the Houston data by MSFMCS using SVM, CEM-MAP, and OTVCA-SVM as the SC in IM, respectively, where MSFMCS was run five times and the results were the best cases among the five different runs with OA. Tables XI–XIII also tabulate P_A, P_{PR}, P_{AA}, P_{OA}, P_{APR}, P_{OPR}, computing times, and AL of running MSFMCS five times using SVM, CEM-MAP, and OTVCA-SVM, respectively, as the SC in the

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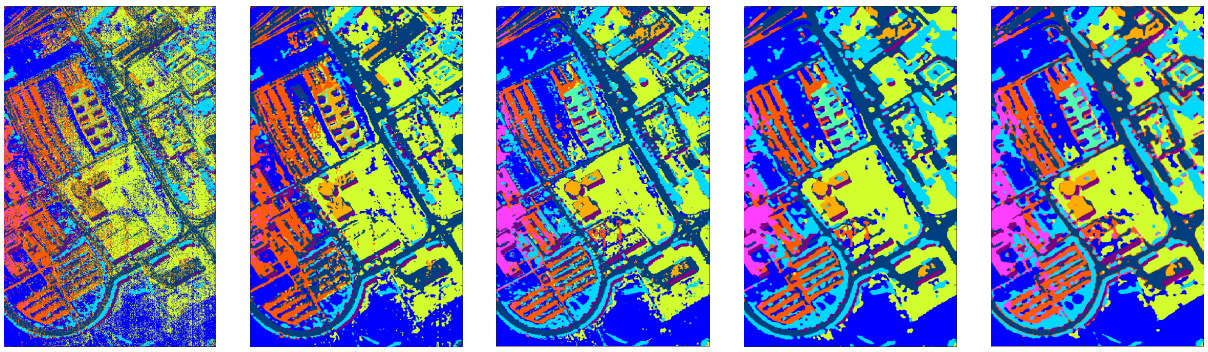


Fig. 11. Classification maps with OA for University of Pavia produced by MSFMCS using CEM as the SC in the IM. (a) CEM-MAP = 69.88%. (b) CEM-Gaussian-MAP = 80.12%. (c) First SFM = 90.86%. (d) Terminal SFM (FTS) = 92.74%. (e) Terminal SFM (RTS) = 93.76%.

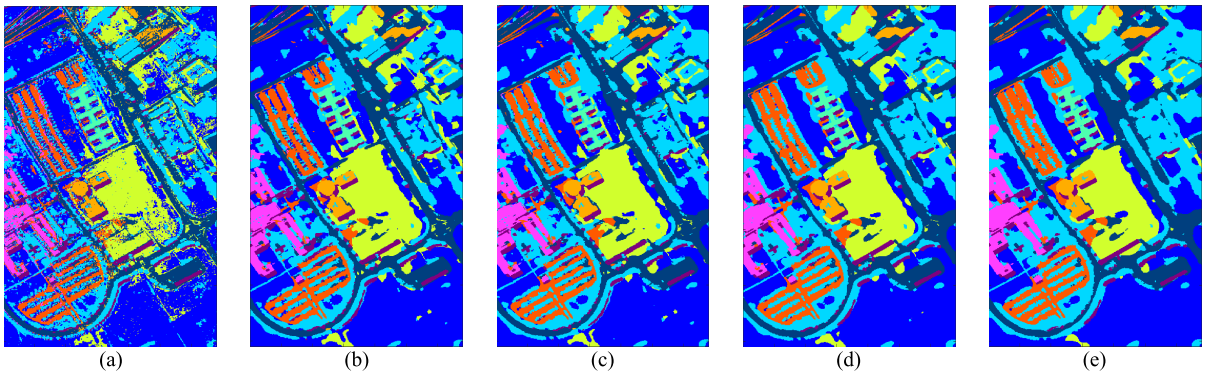


Fig. 12. Classification maps with OA for University of Pavia produced by MSFMCS using OTVCA as the SC in the IM. (a) OTVCA-SVM = 93.91%. (b) OTVCA-EPF-G-g = 97.20%. (c) First SFM = 97.18%. (d) Terminal SFM (FTS) = 97.42%. (e) Terminal SFM (RTS) = 97.92%.

TABLE VIII

P_A, P_{PR}, P_{AA}, P_{OA}, P_{APR}, P_{OPR}, COMPUTING TIMES, AND AL OF MSFMCS USING SVM WITH FIVE RUNS IN THE IM, FIRST SFM, AND TERMINAL SFM WITH 1% TRAINING SAMPLES USING KM FOR UNIVERSITY OF PAVIA

Class	IM SVM		1st SFM EPF-G-g		1st SFM MSFMCS		Terminal SFM(FTS) MSFMCS AL = 4.60±2.19 modules		Terminal SFM(RTS) MSFMCS AL = 4.00±1.22 modules	
	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$
1	79.45±5.05	17.93±1.80	92.19±4.15	18.51±1.85	93.73±1.95	18.49±2.47	94.47±2.51	18.94±2.23	96.07±2.17	18.46±1.84
2	84.48±1.74	27.97±2.54	93.43±2.91	31.88±4.40	91.25±3.70	39.58±6.45	94.06±2.32	38.15±7.99	95.20±3.25	37.17±7.13
3	80.76±4.52	21.43±2.69	93.53±4.88	32.63±3.44	95.35±3.17	31.41±3.55	96.40±3.40	30.36±3.68	96.20±3.15	29.39±3.43
4	93.56±1.75	10.70±1.06	91.44±2.92	10.22±0.96	96.80±0.86	6.06±1.19	94.83±1.73	5.48±0.95	94.12±1.18	5.38±0.66
5	99.28±0.15	33.24±8.18	99.98±0.03	36.31±13.10	99.86±0.13	39.01±12.30	99.58±0.24	37.91±11.60	99.83±0.03	33.73±15.60
6	84.71±4.04	9.22±1.22	97.63±3.93	11.46±3.22	98.20±3.04	13.65±2.83	98.43±2.68	16.07±2.65	99.10±0.82	19.23±5.23
7	91.74±2.40	23.09±4.20	99.57±0.12	37.48±10.60	98.88±1.01	40.82±7.78	99.93±0.13	39.09±9.25	99.98±0.03	41.28±7.43
8	81.01±4.68	17.72±1.74	95.43±3.84	19.32±0.57	95.11±3.45	19.07±1.90	95.81±3.13	19.25±1.74	97.04±1.14	19.91±1.36
9	99.93±0.05	7.58±0.46	100±0	7.10±0.55	99.97±0.04	10.57±1.75	99.66±0.38	12.11±1.74	99.87±0.18	11.67±1.09
P_{AA}	88.32±0.28		95.91±0.86		96.57±0.73		97.02±0.55		**97.49±0.62**	
P_{OA}	84.93±0.96		94.31±1.69		94.08±1.73		95.44±1.21		**96.32±1.60**	
P_{APR}	18.76±0.85		22.76±1.18		**24.30±1.19**		24.15±1.53		24.02±0.96	
P_{OPR}	17.51±0.19		19.45±0.34		19.40±0.35		19.68±0.25		**19.86±0.33**	
Time	19.43±0.43		20.11±0.43		31.69±1.43		73.29±24.50		65.64±11.90	

IM, first SFM, and terminal SFM with 5% training samples using KM for the Houston data where the best results are boldfaced. Two interesting findings were noteworthy. One is that all the three SCs, SVM, CEM-MAP, and OTVCA-SVM, improved their OA performances by 3% (OTVCA-SVM), 7% (CEM-MAP), and 15% (SVM) using EPF-G-g in MSFMCS's first SFM. However, SVM and CEM-MAP really took advantage of MSFMCS's first SFM to improve OA over their corresponding EPF-G-g counterparts used in its first SFM. In particular, CEM-MAP improved nearly 27% compared with only 5% for SVM and 0.5% for OTVCA. At the end, MSFMCS managed to improve the OA of SVM, CEM-MAP, and OTVCA-SVM by 23%, 40%, and 5%, respectively, after the system was terminated. A second one is that the effectiveness of SC used in the IM had tremendous impact on the improvement by MSFMCS. It was particularly evidential when we compared CEM-MAP in Table XII with OTVCA in Table XIII.

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TABLE IX

P_A, P_{PR}, P_{AA}, P_{OA}, P_{APR}, P_{OPR}, COMPUTING TIMES, AND AL OF MSFMCS USING CEM WITH FIVE RUNS IN THE IM, FIRST SFM, AND TERMINAL SFM WITH 1% TRAINING SAMPLES USING KM FOR UNIVERSITY OF PAVIA

Class	IM CEM-MAP		1st SFM CEM-Gaussian-MAP		1st SFM MSFMCS		Terminal SFM(FTS) MSFMCS AL = 16.40±5.94 modules		Terminal SFM(RTS) MSFMCS AL = 5.80±1.78 modules	
	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$
1	78.04	12.31	94.13	12.99	90.73±1.99	16.53±1.13	92.80±3.44	17.20±0.74	93.49±2.16	17.73±0.45
2	72.28	29.83	83.24	32.82	88.71±3.53	32.29±4.66	91.37±3.29	33.85±4.73	92.07±2.38	32.45±3.64
3	24.11	16.16	13.29	25.98	81.12±7.02	21.43±1.95	85.98±8.36	21.53±4.29	88.16±4.72	19.54±3.84
4	78.43	16.32	80.19	15.98	95.16±2.49	9.80±1.75	95.55±2.78	8.84±1.37	94.39±2.40	9.25±1.56
5	25.35	72.40	18.36	72.01	99.92±0.05	50.95±17.0	99.94±0.06	49.17±15.10	99.94±0.06	51.54±13.50
6	78.11	7.63	91.67	8.50	94.40±1.72	10.49±1.05	97.58±1.33	11.38±1.11	96.53±0.60	11.95±0.65
7	63.23	5.66	87.14	15.46	98.39±0.45	26.48±6.58	99.02±0.79	30.92±7.74	99.39±0.40	29.02±8.05
8	68.63	11.58	81.15	12.96	82.84±7.81	20.54±5.06	88.00±6.48	20.33±6.22	88.18±3.80	17.73±4.13
9	72.86	5.04	80.89	7.31	99.13±0.42	9.13±0.98	99.23±0.41	9.93±0.66	99.51±0.33	9.97±0.72
P_{AA}	62.34		70.01		92.27±0.37		94.39±0.65		**94.63±0.63**	
P_{OA}	69.88		80.12		90.16±1.47		92.75±1.63		**93.09±1.13**	
P_{APR}	19.66		22.67		21.96±2.16		**22.57±1.94**		22.13±1.38	
P_{OPR}	14.41		16.52		18.59±0.30		19.13±0.33		**19.20±0.23**	
Time	8.68		9.01		5.72±0.41		123.03±48.9		40.98±12.48	

TABLE X

P_A, P_{PR}, P_{AA}, P_{OA}, P_{APR}, P_{OPR}, COMPUTING TIMES, AND AL OF MSFMCS USING OTVCA WITH FIVE RUNS IN THE IM, FIRST SFM, AND TERMINAL SFM WITH 1% TRAINING SAMPLES USING KM FOR UNIVERSITY OF PAVIA

Class	IM OTVCA-SVM		1st SFM OTVCA-EPF-G-g		1st SFM MSFMCS		Terminal SFM(FTS) MSFMCS AL = 3.00±1.00 modules		Terminal SFM(RTS) MSFMCS AL = 3.00±0.71 modules	
	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$
1	89.43±1.69	16.46±1.23	95.78±2.07	17.27±1.28	96.51±1.87	15.89±3.46	97.12±1.85	16.30±3.39	97.24±2.15	16.44±3.25
2	93.07±1.32	30.73±3.17	96.67±1.85	32.60±4.06	95.66±2.45	35.47±4.44	96.40±2.55	37.76±5.21	97.49±1.48	37.42±4.74
3	87.49±1.98	25.99±1.37	93.47±4.69	29.32±2.65	94.88±4.28	30.84±3.15	96.03±3.62	30.42±2.60	95.79±4.12	28.65±2.82
4	97.31±1.80	8.22±3.10	94.21±3.58	8.08±3.27	96.24±2.79	6.18±1.40	94.80±2.76	5.42±0.95	95.22±0.75	5.16±0.48
5	98.76±1.09	63.11±7.14	99.85±0.21	57.16±4.72	99.62±0.28	58.68±5.31	99.25±0.60	56.75±3.74	99.77±0.10	53.84±2.69
6	98.39±0.71	13.72±2.19	99.82±0.40	14.13±2.85	99.54±0.62	20.36±8.28	99.72±0.44	21.19±7.51	99.74±0.43	23.12±8.06
7	97.33±1.25	30.74±4.88	99.71±0.20	34.99±5.08	99.48±0.32	39.58±11.00	99.84±0.11	38.83±11.80	99.86±0.18	37.63±11.10
8	88.21±3.52	22.71±3.30	94.25±3.74	22.74±4.31	93.26±4.91	21.45±3.32	93.83±4.93	21.32±3.40	93.56±5.00	21.58±1.35
9	99.00±0.60	9.77±1.22	99.83±0.17	10.18±1.47	99.66±0.32	11.89±2.92	99.59±0.34	12.59±1.96	99.42±0.45	12.86±2.13
P_{AA}	94.33±0.70		97.06±1.01		97.21±0.95		97.40±0.95		**97.57±0.98**	
P_{OA}	93.19±0.61		96.63±0.76		96.38±0.87		96.82±0.73		**97.32±0.53**	
P_{APR}	24.61±1.80		25.16±1.67		26.70±2.82		**26.73±2.51**		26.30±2.16	
P_{OPR}	19.22±0.12		19.92±0.15		19.87±0.18		19.96±0.15		**20.07±0.10**	
Time	7.75±0.32		8.46±0.35		19.55±1.30		43.52±15.36		40.05±10.99	

D. Concluding Remarks

Several concluding remarks on MSFMCS are noteworthy.

1) When MSFMCS is implemented for CEM, the used SFs are Gaussian filters. In addition, to fairly compare CEM with EPF-G-g, the original Otsu's method implemented in CEM in [15]–[17] was replaced by MAP which was used by EPF-G-g.
2) All the above experiments demonstrated that SVM, CEM-MAP, and OTVCA-SVM implemented in IM of MSFMCS can be significantly improved by the first SFM.
3) Among all the three SCs considered in IM of MSFMCS, OTVCA-SVM performed the best, while CEM-MAP was the worst. This made sense because OTVCA is a feature extraction technique followed by SVM. So OTVCA-SVM certainly outperformed SVM alone. On the other hand, CEM did not do well as SVM and OTVCA-SVM did. This is mainly because CEM is a linear filter compared with SVM which is a nonlinear kernel-based classifier.
4) SVM, CEM-MAP, and OTVCA-SVM used in IM can be implemented in two versions in the first SFM. One is to use MAP to terminate MSFMCS at the first SFM as SVM-EPF-MAP which turns out to be EPF-G-g in [7], CEM-Gaussian-MAP, and OTVCA-EPF-G-g with their results tabulated in the second column in Tables V–XIII. The other is to use SVM as a stopping rule to determine whether MSFMCS is continued to cascade a new SFM for next round processing as SVM-EPF-SVM, CEM-Gaussian-SVM, and OTVCA-SVM-EPF-SVM with their results tabulated in the third column in Tables V–XIII. In both cases, the classification of the SC used in IM can be significantly improved by the first SFM as shown by experiments for all the three studied image scenes.
5) The classification performance of the first SFM can be further improved by subsequent cascaded SFMs according to the conducted experiments. However, the degree of improvement is heavily determined by the effectiveness of the original SC used in IM. If an SC is more effective, then the room of its improvement by

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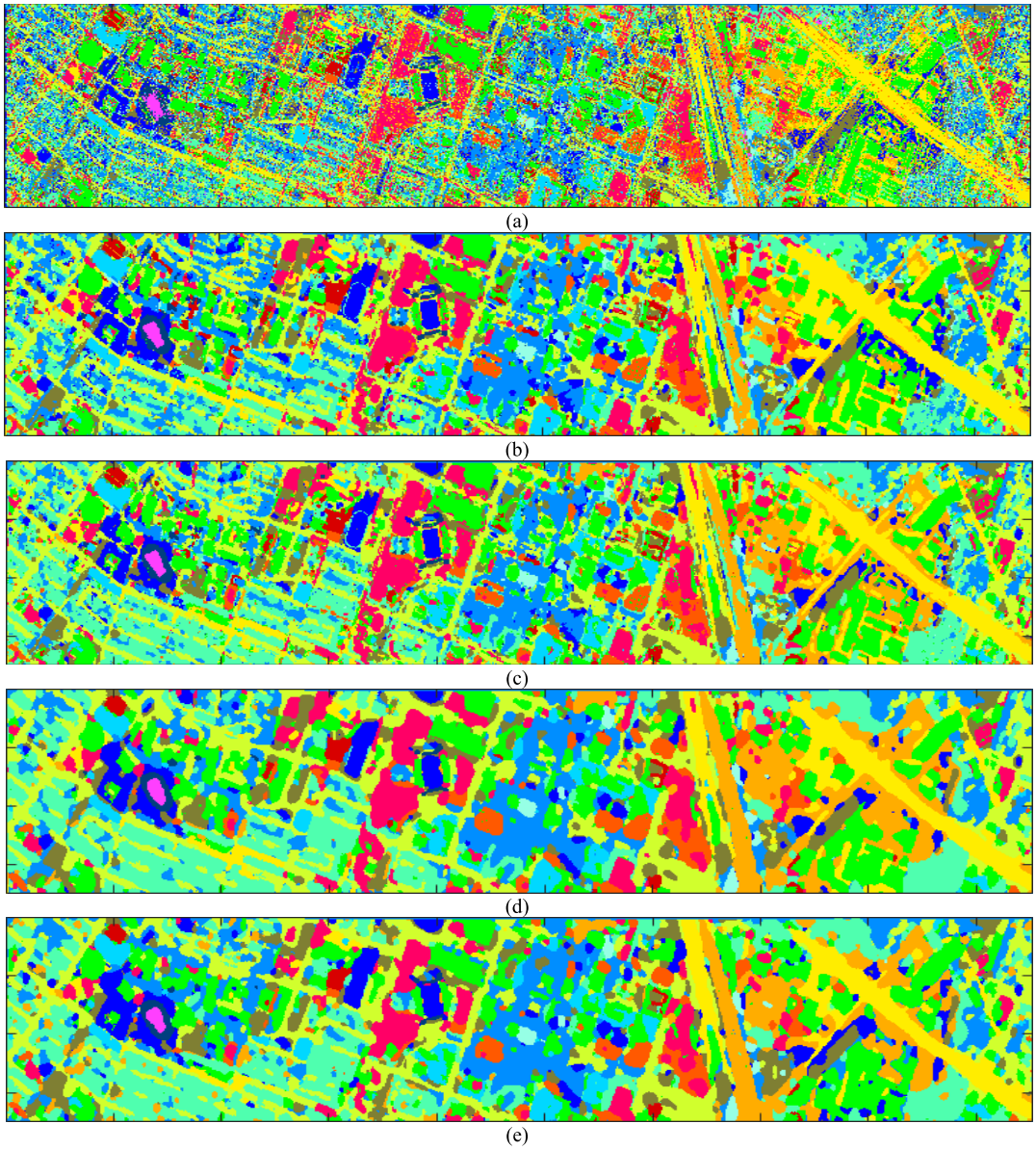


Fig. 13. Classification maps with OA for the Houston data produced by MSFMCS using SVM as the SC in the IM. (a) SVM = 66.86%. (b) EPF-G-g = 81.82%. (c) First SFM = 86.98%. (d) Terminal SFM (FTS) = 88.35%. (e) Terminal SFM (RTS) = 89.83%.

MSFMCS will be small. Otherwise, the improvement will be significant and greater if the used SC is less effective.

6) Upon implementing CEM, the class means were used as the desired signatures **d** and training samples were selected to implement the SVM-based stopping rule. In other words, there is no training phase for CEM but only SVM used as a stopping rule requires training samples.
7) One important and crucial difference between MSFMCS and the existing HSIC methods is its use of multiple SFMs. In the context of MSFMCS, any classifier used for HSIC can be considered the SC operated in IM such as SVM, OTVCA-SVM, and CEM-MAP. In MSFMCS, it is the SVM-based stopping rule attached to the end of each SFM that is the key to determine whether the cascaded system should be terminated.
8) According to Tables XI–XIII, P_{OPR} for the Houston data was very low around 1%–2% for all the three SCs compared with 36%–47% for the Purdue data and 14%–20% for U. of Pavia even though the P_{OA} values of these three scenes could be very high at least 84% up to 97%. This suggested that the urban BKG of the Houston data is much more complicated than the agriculture BKG of the Purdue data and the campus BKG of U. of Pavia. As a result, little improvement could be gained by MSFMCS due to the fact that the Houston data have more complicated BKG than the Purdue BKG and U. of Pavia BKG. However, it should be noted that when a data sample is referred to as a

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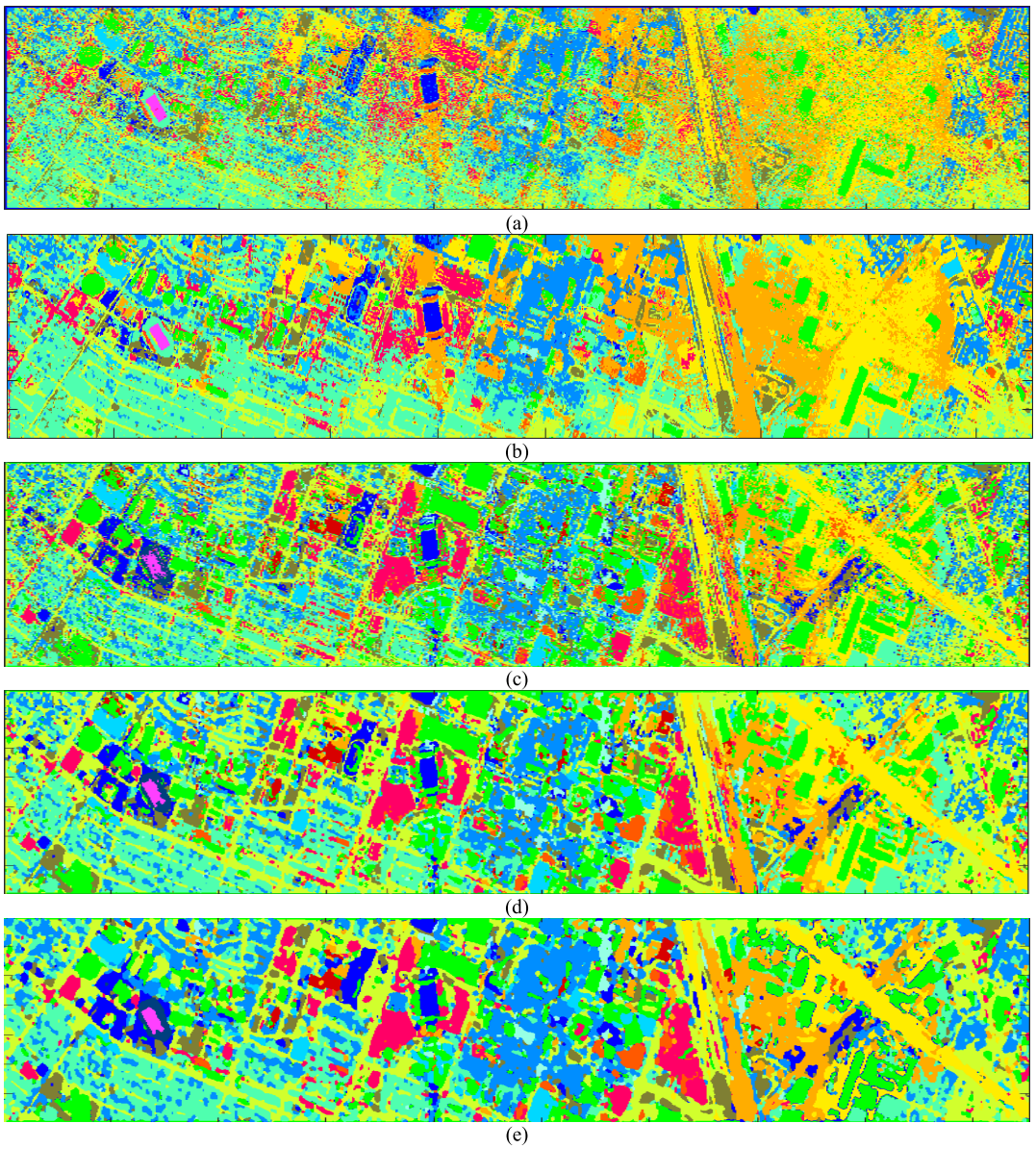


Fig. 14. Classification maps with OA for the Houston data produced by MSFMCS using CEM as the SC in the IM. (a) CEM-MAP = 48.76%. (b) CEM-Gaussian-MAP = 55.83%. (c) First SFM = 81.35%. (d) Terminal SFM (FTS) = 87.28%. (e) Terminal SFM (RTS) = 91.41%.

BKG data sample, it does not imply that this data sample is of no interest or not important but rather indicates that it is a data sample which cannot be labeled by ground truth. In many cases, such data samples are more important and crucial than ground-truth-labeled data samples, specifically the Houston urban scene where many interesting objects cannot be truthfully labeled by ground truth. Unfortunately, the BKG issue has not received much attention in classification as it should. PR is probably the only measure that can be used for this purpose.

9) Finally, we have also done experiments with various percentages of training samples such as 10% and 1% for the Purdue data and 2% and 0.5% for U. of Pavia, and the conclusions were similar. Due to limited space, these experiments are not included here.

VIII. Generalizability of MSFMCS

MSFMCS has its great generalizability potential in many future research directions. There are several future research directions which are currently investigated and undertaken.

1) *IM*: In MSFMCS, the IM implements a selected SC. However, this SC can be extended to any hyperspectral classifier regardless of whether or not it is an SS classifier. This is because the SC implemented in MSFMCS is independent of subsequent SFMs. It can accommodate more different types of heterogeneous classifiers that can be used in IM of MSFMCS, such as spectral MRF-based classifiers [5], spatial–spectral-based classifiers 51], joint SS-based classifiers [3], [4], [51], and CNN-based classifiers as initial classifiers, all of which can be used to generalize MSFMCS. As a result, all these classifiers

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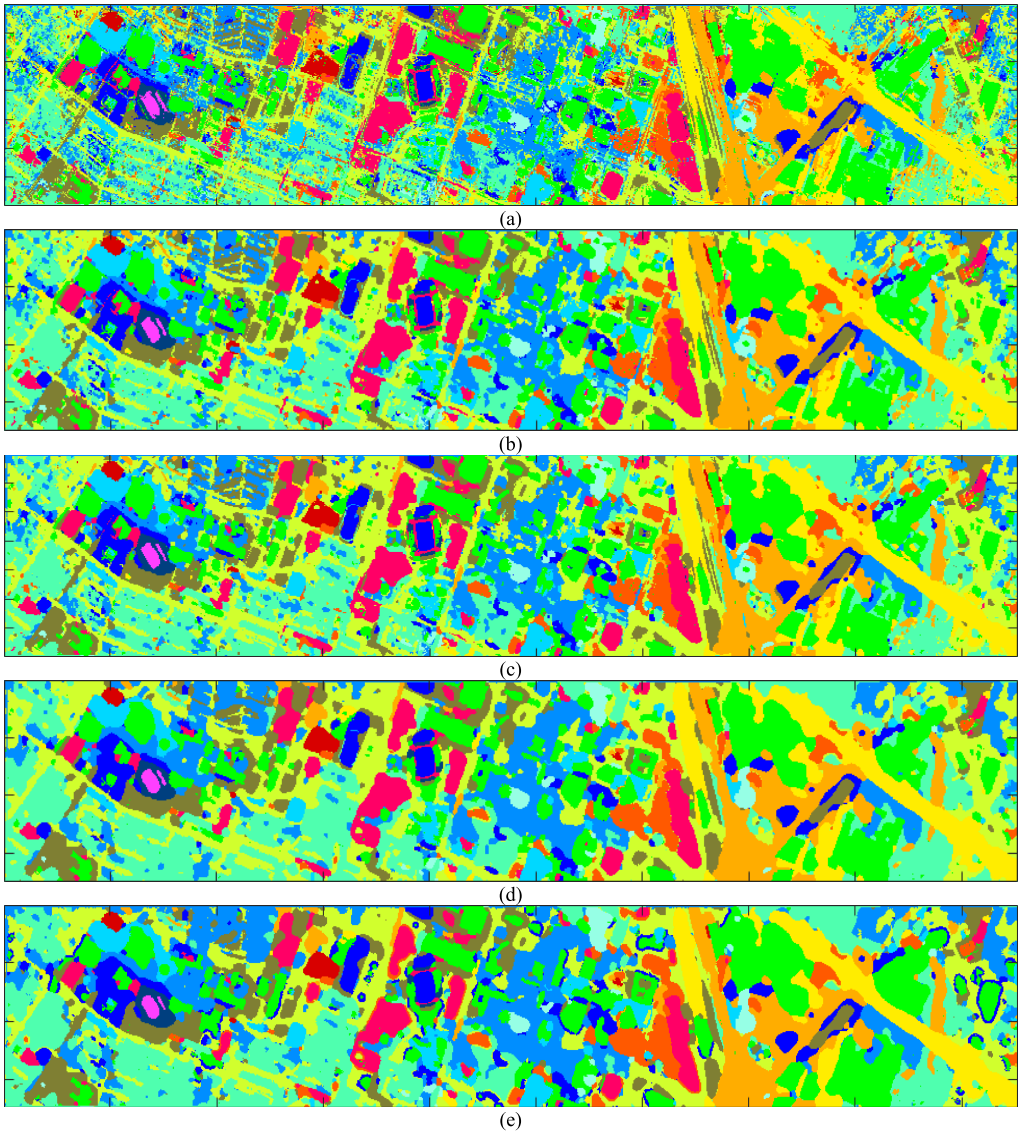


Fig. 15. Classification maps with OA for the Houston data produced by MSFMCS using OTVCA as the SC in the IM. (a) OTVCA-SVM = 88.52%. (b) OTVCA-EPF-G-g = 91.98%. (c) First SFM = 91.93%. (d) Terminal SFM (FTS) = 92.65%. (e) Terminal SFM (RTS) = 92.96%.

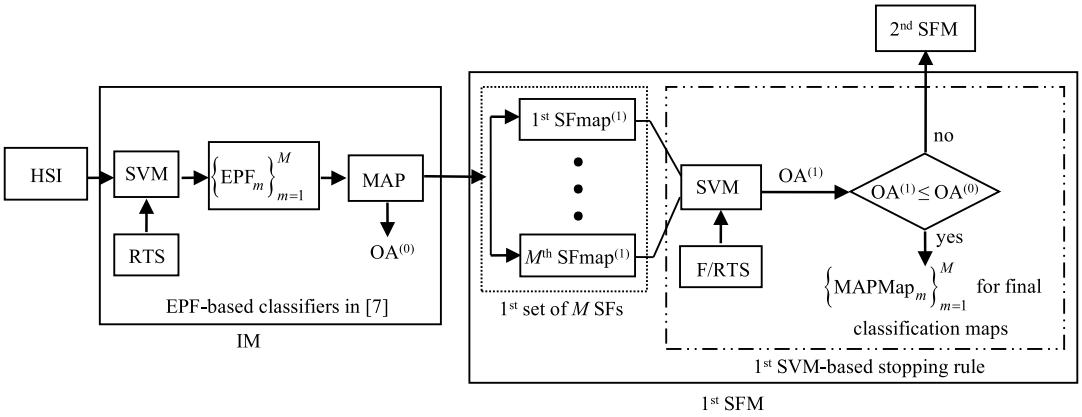


Fig. 16. Block diagram of implementing EPF-based classifiers by MSFMCS as the first SFM.

implemented in IM as an input module can be further improved by subsequent SFMs in MSFMCS.

Following the above generalizability, the EPF-based classifiers developed in [7] can also be implemented in an IM instead of the first SFM as proposed in Fig. 2(a). To be specific, Fig. 2(a) can be further modified as shown in Fig. 16 where the MAP classification maps, $\{\text{MAPMap}_m\}_{m=1}^{M}$, directly resulting from the

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TABLE XI

P_A, P_{PR}, P_{AA}, P_{OA}, P_{APR}, P_{OPR}, COMPUTING TIMES, AND AL OF MSFMCS USING SVM WITH FIVE RUNS IN THE IM, FIRST SFM, AND TERMINAL SFM WITH 5% TRAINING SAMPLES USING KM FOR HOUSTON DATA

Class	IM SVM		1st SFM EPF-G-g		1st SFM MSFMCS		Terminal SFM(FTS) MSFMCS AL = 3.40±1.67 modules		Terminal SFM(RTS) MSFMCS AL = 3.40±1.34 modules	
	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$
1	77.44±5.28	2.16±0.43	86.63±7.14	2.67±0.63	88.34±2.58	3.43±1.08	89.12±3.12	3.62±1.15	90.43±2.77	3.53±0.76
2	68.30±3.57	1.49±0.13	79.96±4.29	2.18±0.30	84.88±2.96	2.78±0.15	84.51±1.81	2.77±0.14	86.93±1.41	2.69±0.25
3	83.47±5.21	10.48±4.69	89.29±7.16	29.01±14.80	95.20±2.02	50.73±10.90	92.48±3.23	58.96±3.35	92.96±2.37	57.61±5.11
4	57.81±3.90	0.89±0.06	78.68±1.75	0.96±0.10	80.16±2.39	1.11±0.11	81.22±1.83	1.10±0.13	80.96±2.95	1.20±0.06
5	87.61±0.70	3.51±0.47	96.45±0.56	4.35±0.59	96.07±1.72	6.00±1.66	96.95±1.66	6.28±1.35	98.93±0.74	6.05±1.53
6	71.87±4.76	0.98±0.28	90.40±6.06	3.97±2.06	94.95±3.18	4.57±1.25	96±2.06	6.34±2.00	96.43±1.97	5.85±1.85
7	44.63±2.88	0.81±0.05	69.92±4.54	1.17±0.12	82.08±1.60	0.95±0.10	84.92±3.15	0.98±0.11	86.52±2.74	0.99±0.08
8	64.91±2.53	1.08±0.11	70.61±3.09	1.16±0.13	75.98±1.63	1.11±0.17	77.13±2.07	1.13±0.15	80.01±2.57	1.07±0.18
9	54.05±3.44	0.82±0.04	80.86±4.20	0.88±0.06	78.64±6.01	1.00±0.14	82.77±3.03	0.98±0.15	84.68±3.89	1.00±0.15
10	73.88±2.15	1.62±0.18	87.90±5.70	1.98±0.28	94.58±3.67	2.30±0.48	95.38±3.16	2.47±0.55	97.01±2.42	2.70±0.53
11	54.78±5.16	1.83±0.37	70.89±6.51	3.34±1.02	84.72±5.67	2.79±1.12	88.61±5.80	3.06±1.50	92.29±4.42	2.78±1.05
12	70.98±4.28	1.92±0.15	84.07±5.02	2.17±0.17	90.12±1.92	2.55±0.30	91.71±1.02	2.63±0.31	92.03±2.43	2.83±0.59
13	56.11±5.56	1.03±0.05	84.22±7.77	1.88±0.16	91.98±4.76	1.74±0.23	94.28±4.09	1.91±0.34	95.99±4.45	1.93±0.33
14	87.57±1.99	3.70±1.13	97.42±1.00	7.45±3.69	98.73±0.80	8.18±2.64	100±0	10.46±2.22	100±0	12.93±3.27
15	68.90±4.66	5.33±2.25	81.42±8.06	14.00±4.87	90.60±3.69	14.83±4.19	90.00±4.56	16.42±3.99	88.87±3.91	16.25±5.41
P_{AA}	68.15±1.19		83.25±1.56		88.47±0.89		89.67±1.14		**90.94±0.87**	
P_{OA}	66.86±0.94		81.82±0.98		86.98±1.07		88.35±1.20		**89.83±0.94**	
P_{APR}	2.51±0.31		5.15±0.96		6.94±0.85		7.94±0.34		**7.96±0.73**	
P_{OPR}	1.51±0.02		1.85±0.02		1.97±0.02		2.00±0.02		**2.03±0.02**	
Time	162.68±1.50		168.04±1.54		226.15±1.18		372.26±99.56		380.71±89.73	

TABLE XII

P_A, P_{PR}, P_{AA}, P_{OA}, P_{APR}, P_{OPR}, COMPUTING TIMES, AND AL OF MSFMCS USING CEM WITH FIVE RUNS IN THE IM, FIRST SFM, AND TERMINAL SFM WITH 5% TRAINING SAMPLES USING KM FOR HOUSTON DATA

Class	IM CEM-MAP		1st SFM CEM-Gaussian-MAP		1st SFM MSFMCS		Terminal SFM(FTS) MSFMCS AL = 12.00±1.87 modules		Terminal SFM(RTS) MSFMCS AL = 9.40±3.71 modules	
	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$
1	46.04	2.62	50.44	5.04	82.43±3.04	2.92±0.79	85.72±3.43	3.29±1.34	87.88±3.84	3.39±0.60
2	52.87	1.74	57.34	2.14	78.40±3.77	1.99±0.18	81.69±3.19	2.46±0.27	84.13±2.95	2.61±0.59
3	66.43	48.79	63.70	51.51	90.18±2.70	45.96±12.20	94.49±2.70	55.23±13.10	93.85±5.53	60.63±6.39
4	52.57	0.88	57.56	0.89	71.49±4.15	1.08±0.05	75.56±3.27	1.13±0.10	81.43±4.14	1.11±0.10
5	59.66	10.78	65.22	13.77	94.87±1.75	5.43±1.20	96.11±1.09	6.83±1.54	97.72±1.13	5.67±0.92
6	41.54	33.75	46.15	34.17	92.61±4.62	2.15±0.43	95.26±4.52	3.28±0.62	95.50±2.74	4.72±1.51
7	64.43	0.52	76.66	0.58	67.77±3.19	0.95±0.12	77.53±4.06	0.96±0.14	78.81±3.25	0.98±0.07
8	35.93	1.33	36.17	1.50	78.79±5.04	1.04±0.08	82.66±5.71	1.08±0.02	86.09±3.89	1.12±0.09
9	52.80	0.65	62.46	0.77	71.51±3.14	0.92±0.10	81.72±5.68	0.89±0.13	84.82±2.44	0.92±0.08
10	57.70	0.83	73.76	0.99	91.78±2.99	2.07±0.14	94.37±2.81	2.30±0.14	96.18±1.89	2.47±0.34
11	65.51	0.77	83.08	0.99	82.12±7.05	2.12±0.30	88.21±5.67	2.33±0.35	92.43±1.61	2.19±0.30
12	28.87	1.07	34.63	1.32	89.60±3.37	2.91±0.29	91.59±1.93	3.39±0.28	94.38±2.89	3.59±0.35
13	31.13	1.89	50.11	4.93	84.09±5.55	2.35±0.06	92.32±5.75	3.00±0.75	92.57±5.51	4.13±1.28
14	35.51	63.33	28.74	63.73	97.66±1.44	5.69±1.27	99.43±0.74	10.86±3.56	99.90±0.20	12.34±2.11
15	0.00	0.00	0.00	0.00	83.27±5.64	14.99±2.86	90.33±7.75	21.06±3.50	93.48±2.94	21.26±3.53
P_{AA}	46.07		52.40		83.77±0.68		88.47±1.16		**90.61±1.37**	
P_{OA}	48.76		55.83		82.20±0.87		86.93±1.10		**89.44±1.20**	
P_{APR}	11.26		12.16		6.17±1.01		7.88±1.22		**8.48±0.76**	
P_{OPR}	1.10		1.26		1.86±0.01		1.97±0.02		**2.02±0.02**	
Time	30.79		32.64		41.38±2.43		394.24±66.72		302.14±119.45	

EPF-based classifier in [7] can be considered as $OA^{(0)}$ in the IM and $OA^{(1)}$ is produced in the first SFM of MSFMCS.

2) *Model Expansion*: According to Fig. 2(b), if we assume the IM as an initial condition, $\mathbf{Y}_k$ is the kth $\left\{\text{SVMap}_m^{(k)}\right\}_{m=1}^{M}$ produced by the kth SVM-based stopping rule, and $\mathbf{X}_k$ is the kth SF-filtered real-valued classification maps, $\left\{\text{SFMap}_m^{(k)}\right\}_{m=1}^{M}$, MSFMCS can be realized as an autoregressive (AR) filter by $\mathbf{Y}_k = \mathbf{Y}_{k-1} \oplus \mathbf{X}_k$ where $\oplus$ denotes a cascade connector in Fig. 3 [52]. On the other hand, the ISS classifiers developed in [12] and [14]–[17] can be considered as moving averaging (MA) filter by $\mathbf{Y}_k = \mathbf{X}_k \hat{+} \mathbf{X}_{k-1}$ where $\hat{+}$ is an additive operation. It is interesting to take advantages of both the structures by integrating these two models into an auto-regressive moving averaging (ARMA) model which implements one forward system and one backward system.

3) *Target/Anomaly Detection*: MSFMCS offers a paradigm for a potential application in target and anomaly detection since CEM is shown to be effective in classification by experiments in Section VII.

4) *Fusion*: MSFMCS provides a framework to fuse two or more classifiers. The key idea is derived from [53] by introducing a fusion process sandwiched between SFMaps and SVM-based stopping rules module by module.

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TABLE XIII

P_A, P_{PR}, P_{AA}, P_{OA}, P_{APR}, P_{OPR}, COMPUTING TIMES, AND AL OF MSFMCS USING OTVCA WITH FIVE RUNS IN THE IM, FIRST SFM, AND TERMINAL SFM WITH 5% TRAINING SAMPLES USING KM FOR HOUSTON DATA

Class	IM OTVCA-SVM		1st SFM OTVCA-EPF-G-g		1st SFM MSFMCS		Terminal SFM(FTS) MSFMCS AL = 2.40±1.14 modules		Terminal SFM(RTS) MSFMCS AL = 3.40±1.34 modules	
	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$	$P_A(C_m)$	$P_{PR}(C_m)$
1	89.76±2.77	3.44±0.88	91.33±3.97	3.86±1.00	91.63±4.12	4.06±0.94	91.63±3.95	4.07±0.93	91.54±4.54	3.93±1.02
2	84.00±4.65	2.04±0.26	86.81±5.66	2.29±0.34	86.60±4.89	2.35±0.26	86.95±5.93	2.37±0.24	88.77±5.56	2.33±0.22
3	95.40±1.14	54.93±5.33	97.47±1.49	57.53±7.51	97.79±1.26	57.26±3.10	96.95±1.76	56.92±1.75	93.97±3.13	60.28±3.54
4	78.21±2.50	1.03±0.12	84.38±2.36	1.08±0.16	83.97±1.93	1.13±0.17	85.06±1.00	1.09±0.16	84.29±2.51	1.15±0.18
5	95.97±2.83	5.72±0.67	97.13±2.91	6.18±0.78	97.19±2.94	6.79±1.08	97.89±1.74	6.89±0.83	98.37±1.35	6.81±1.05
6	95.44±6.19	3.89±2.30	96.86±4.86	4.56±2.70	96.92±5.20	4.57±2.41	97.41±3.75	4.53±2.25	98.21±1.09	4.70±1.87
7	71.21±3.08	0.86±0.08	76.81±4.92	0.89±0.10	78.94±4.88	0.82±0.09	80.64±4.88	0.82±0.09	82.11±5.30	0.84±0.09
8	84.05±5.61	1.12±0.07	85.85±4.86	1.15±0.09	86.49±5.65	1.16±0.11	86.54±5.60	1.15±0.12	89.51±1.47	1.12±0.08
9	77.14±1.64	0.97±0.08	84.21±1.78	1.02±0.08	85.25±1.92	1.01±0.13	85±1.58	1.04±0.14	87.02±2.00	1.02±0.04
10	96.46±1.08	2.69±0.30	98.71±0.87	2.74±0.34	98.32±1.14	2.97±0.40	98.40±1.23	3.04±0.41	98.81±1.18	3.21±0.11
11	92.61±2.34	2.81±0.19	95.88±3.61	3.00±0.18	95.56±3.60	3.21±0.31	96.04±3.88	3.31±0.30	96.64±3.19	3.28±0.32
12	94.19±1.52	3.29±0.37	96.15±1.05	3.26±0.44	96.23±1.20	3.33±0.35	96.41±0.63	3.36±0.38	96.23±0.78	3.66±0.49
13	86.01±1.70	1.96±0.09	93.13±1.45	2.32±0.12	94.28±1.71	2.25±0.13	95.52±2.78	2.34±0.29	96.54±2.81	2.32±0.29
14	98.87±0.94	9.89±2.05	100±0	11.60±2.86	100±0	13.23±3.59	100±0	14.15±3.54	100±0	15.77±3.48
15	93.96±2.53	18.20±0.85	96.03±3.94	20.18±1.36	97.21±2.12	21.61±2.49	95.27±2.06	21.82±3.22	94.69±2.05	21.22±3.65
P_{AA}	88.89±1.07		92.05±1.04		92.42±1.01		92.65±1.02		**93.11±0.80**	
P_{OA}	87.60±0.84		90.87±0.91		91.22±0.88		91.51±0.99		**92.12±0.84**	
P_{APR}	7.52±0.63		8.11±0.88		8.38±0.74		8.46±0.74		**8.78±0.48**	
P_{OPR}	1.98±0.01		2.05±0.02		2.06±0.02		2.07±0.02		**2.08±0.01**	
Time	59.20±1.53		63.57±1.53		115.50±12.42		256.59±72.48		310.65±90.11	

IX. CONCLUSION

This article presents a general framework for HSIC, called MSFMCS, which has a simple architecture with no model complexity and is very easy to implement. Several contributions can be summarized as follows.

1) MSFMCS extends any SC classifier by including a cascade of SFMs so that its classification performance can be improved as MSFMCS progresses SFM by SFM.
2) More interestingly, many SS classifiers can be considered to perform spectral classification in IM followed by its SFs implemented in the first SFM so that such SS classifier can be further improved by MSFMCS via a cascade of SFMs.
3) MSFMCS includes SFMs to capture spatial class information that can be extracted from the preceding SFM and passed on to subsequent SFMs module by module to increase classification accuracy.
4) MSFMCS designs an automatic SVM-based stopping rule at the end of each SFM as shown in Fig. 2 to determine whether or not the system needs to be continued and terminated. In this case, the length, that is, the number of SFMs, is automatically determined without prior knowledge.
5) MSFMCS only uses training samples to train the SC implemented in IM and training samples for SVM-based stopping rule. It is quite different from deep learning and feedforward NNs which use training samples to train the network. MSFMCS does not require any system training since it does not use weights to connect between SFMs but rather uses SVM-based stopping rules as a classifier and a decision-maker. In this case, no weight training phase is needed such as backpropagation commonly used in multilayer perceptron.
6) MSFMCS can implement two types of training sampling for SVM, FTS, and RTS at the end of each SFM as opposed to a feedforward NN which itself is a classifier and selects training samples to train interweights connecting the network.

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Xiaodi Shang (Student Member, IEEE) received the B.S. degree in software engineering from Qingdao University, Qingdao, China, in 2016. She is pursuing the Ph.D. degree in computer application technology with Dalian Maritime University, Dalian, China.

Her research interests include hyperspectral image classification, band selection, and applications.

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Meiping Song (Member, IEEE) received the Ph.D. degree from the College of Computer Science and Technology, Harbin Engineering University, Harbin, China, in 2006.

From 2013 to 2014, she was a Visiting Associate Research Scholar with the Remote Sensing Signal and Image Processing Laboratory, University of Maryland, Baltimore County, Baltimore, MD, USA. She has been a Professor with the College of Information Science and Technology, Dalian Maritime University, Dalian, China, since 2020. Her research interests include remote sensing and hyperspectral image processing.



Chein-I Chang (Life Fellow, IEEE) received the B.S. degree in mathematics from Soochow University, Taipei, Taiwan, in 1973, the M.S. degree in mathematics from the Institute of Mathematics, National Tsing Hua University, Hsinchu, Taiwan, in 1975, the M.A. degree in mathematics from the State University of New York at Stony Brook, Stony Brook, NY, USA, in 1977, the M.S. degree in mathematics and the M.S.E.E. degree in electrical engineering from the University of Illinois at Urbana–Champaign, Urbana, IL, USA, both in 1982, and the Ph.D. degree in electrical engineering from the University of Maryland, College Park, MD, USA, in 1987.

He has been with the University of Maryland, Baltimore County (UMBC), Baltimore, MD, since 1987, and is a Professor with the Department of Computer Science and Electrical Engineering. He has been holding the Chang Jiang Scholar Chair Professorship and is the Director of the Center for Hyperspectral Imaging in Remote Sensing (CHIRS), Dalian Maritime University, Dalian, China, since 2016. In addition, he has been a Chair Professor with National Chiao Tung University, Hsinchu, since 2019. He has several patents on hyperspectral image processing. He has authored four books, *Hyperspectral Imaging: Techniques for Spectral Detection and Classification* (Kluwer Academic Publishers, 2003), *Hyperspectral Data Processing: Algorithm Design and Analysis* (John Wiley & Sons, 2013), *Real Time Progressive Hyperspectral Image Processing*: *Endmember Finding and Anomaly Detection* (Springer, 2016), and *Recursive Hyperspectral Sample and Band Processing: Algorithm Architecture and Implementation* (Springer, 2017). In addition, he also edited two books, *Recent Advances in Hyperspectral Signal and Image Processing* (2006) and *Hyperspectral Data Exploitation: Theory and Applications* (John Wiley & Sons, 2007), and co-edited with A. Plaza a book on *High Performance Computing in Remote Sensing* (CRC Press, 2007). His research interests include multispectral/hyperspectral image processing, automatic target recognition, and medical imaging.

Dr. Chang is a fellow of SPIE. He received the National Research Council Senior Research Associateship Award, from 2002 to 2003, sponsored by the U.S. Army Soldier and Biological Chemical Command, Edgewood Chemical and Biological Center, Aberdeen Proving Ground, MD. He is an Associate Editor of *Remote Sensing* and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.

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